

SITE SAFETY PLUS

Construction site health and safety

D: High risk activities



Published by CITB, Sand Martin House, Bittern Way, Peterborough, PE2 8TY

© **Construction Industry Training Board 1982**

First published in 1982 as GE700 (A4 ring binder x2)

Republished in 2013 as GE700D (A4 book)

Revised 2014, 2015

Reprinted Jan 2016, Jan 2017

Revised Dec 2017, Dec 2018, Dec 2019, Dec 2020, 2022

Revised 2023, 2024

ISBN 978-1-85751-584-8

SSP, SMSTS, SSSTS, HSA, DRHS and SEATS are all registered trade marks
of the Construction Industry Training Board

CITB is registered as a charity in England and Wales (Reg No 264289)
and in Scotland (Reg No SC044875)



CITB has made every effort to ensure that the information contained within this publication is accurate. Its content should be used as guidance only and not as a replacement for current regulations, existing standards or as a substitute for legal advice and is presented without any warranty, either express or implied, as to its accuracy. In no event will CITB be liable for any damages arising from reliance upon its content.

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system or transmitted in any form or by any means, electronic, mechanical, photocopying, recording or otherwise, without the prior permission in writing from CITB save that it may be read and saved by the original recipient, for whose private use it was originally intended.

CONTENTS



High risk activities

01	Work at Height Regulations 2005	1
02	Working at height	15
03	Common access equipment	37
04	Scaffolding	51
05	Fall arrest and suspension equipment	75
06	Excavations	91
07	Underground and overhead services	105
08	Confined spaces	117
09	Dangerous substances	131
	Index	153



**Scan the QR code to complete a brief survey,
and share your feedback on the course.**

CONTENTS

Work at Height Regulations 2005

01

1.1	Introduction	2
1.2	Important points	3
1.3	Requirements, interpretation and duties (Regulations 2 and 3)	3
1.4	Organisation and planning of work (Regulation 4)	5
1.5	Competence (Regulation 5)	5
1.6	Avoidance of risks from work at height (Regulation 6)	6
1.7	Selection of equipment for work at height (Regulation 7)	7
1.8	Requirements for particular work equipment (Regulation 8)	7
1.9	Fragile surfaces (Regulation 9)	9
1.10	Falling objects (Regulation 10)	9
1.11	Danger areas (Regulation 11)	9
1.12	Inspection of work equipment (Regulation 12)	9
1.13	Inspection of places of work at height (Regulation 13)	10
1.14	Schedules 1 to 8	10
	Appendix A – Work at height flowchart	14

Supporting INFORMATION

GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.



**Scan the QR code to complete a brief survey,
and share your feedback on the course.**

Overview

Falls while working at height remain the leading cause of fatal accidents and specified injuries in the construction industry. The majority of falls are from heights of less than 2 m.

Many accidents are caused by the poor selection of access equipment, by failures to provide fall prevention measures and by workers not using access equipment in a safe manner.

This chapter outlines what the Work at Height Regulations 2005 are about, what they require, what needs to be done and by whom.

1.1 Introduction

The purpose of the regulations is to reduce the number of deaths, injuries and personal suffering resulting from falls from height, and so improve the safety performance of the industry. The Work at Height Regulations came into force in 2005 and were amended in 2007.

The regulations cover all circumstances where a person is working at height, is gaining access to, or egress from, such a place of work, at, above or below ground level, where persons or objects could fall a distance liable to cause personal injury (that is, any distance whatsoever). These regulations do not apply to fixed staircases.

Common examples of tasks carried out within the construction industry, classified as work at height, are shown below.

- Working on a scaffold.
- Working from a mobile elevating work platform (MEWP).
- Standing on, or working from, the back of a lorry.
- Working close to an excavation's leading edge.
- Working on staging or trestles.
- Using a mobile access tower (MAT).
- Using ladders or stepladders.
- Working in an excavation or shaft.

Many other jobs in the construction industry also involve working at height and are covered by the Work at Height Regulations. Health and Safety Executive (HSE) evaluation reports of accidents that occurred during work at height show that common contributory factors include failures of the following.

- All parties to recognise that there was a problem.
- Management to provide adequate information, instruction, training or supervision
- Management to ensure that the safe system of work was followed.
- Management to provide a safe system of work.
- The injured worker to use appropriate equipment.
- Management to provide safe plant and equipment.
- Collective protection to take priority over personal protection.



Falls from height - 2021-22 statistics

- In 2021/22, 29 fatal injuries were due to falls from a height, accounting for 24% of all worker deaths over the year. While the number of fatal injuries has generally fallen over the past decade, the percentage of fatal injuries has not decreased.



Falling from height

A timber company has been fined £80,000 after an employee fell through a stairwell while working on a barn conversion.

The 49-year-old driver and warehouse operative was working on the barn, which was being converted for rental use in May 2018.

He was trying to access the first floor of the barn from exterior scaffolding and jumped onto a piece of insulation which was covering a stairwell. The insulation gave way and the worker fractured two vertebrae in the fall.

An investigation by the HSE found the work was not properly planned, appropriately supervised, or carried out in a safe manner when the incident occurred. The company had a duty to control how the work was carried out, including staff supervision.

The timber company pleaded guilty to breaching Section 4 (1) of the Work at Height Regulations 2005, and was fined £80,000 and ordered to pay costs of £7,331 at the local Magistrates' Court. The company also ordered to pay a £170 victim surcharge.

Speaking after the hearing, a HSE inspector said: "This worker's injuries were serious. This incident could have been avoided if basic safeguards had been put in place. Falls from height remain one of the most common causes of work-related fatalities and injuries in this country and the risks associated with working at height are well-known."



The HSE website has many resources, including a dedicated work at height section that includes information on the Work at Height Regulations, including *Working at height: a brief guide* (INDG401).

1.2 Important points

- Falls from height are the main cause of fatalities and specified injuries to construction workers.
- Falls from low heights (below 2 m) are the cause of many deaths and injuries.
- Working at height is defined as working at any height from which, if measures are not taken, a fall (of a person, materials or equipment) could occur and cause personal injury.
- Work at height must be carried out in accordance with the Work at Height Regulations, which require that such work is:
 - avoided if it is reasonably practicable to do the job another way
 - carried out using appropriate equipment to prevent falls
 - organised so that the distance and possible consequence of any fall are minimised
 - risk-assessment based
 - properly planned and supervised by a competent person(s)
 - carried out by competent operatives.

Duty holders must:

- avoid work at height where they can
- use work equipment or other measures to prevent falls where they cannot avoid working at height
- where they cannot eliminate the risk of a fall, use work equipment or other measures to minimise the distance and consequences of a fall should one occur

Collective protection for employees as a whole (for example, guard-rails) must take priority over personal protection (safety harnesses) at all times.



Working at height is defined as working at any height from which, if measures are not taken, a fall could occur and cause personal injury.



The Access Industry Forum website contains further information and guidance for work at height activities that can be downloaded upon request.

1.3 Requirements, interpretation and duties (Regulations 2 and 3)

1.3.1 Requirements of the regulations

The regulations require employers to ensure the following.

- Where it is reasonably practicable, the need to carry out work at height is avoided.
- Where such work cannot be avoided, the most appropriate equipment for the work and to prevent falls is selected.
- The distance of, and potential consequences of, any fall are reduced.
- The work is properly planned, including planning for emergencies and rescue. It must be risk-assessment based and carried out safely.
- The work is carried out by trained and competent persons who are adequately supervised.

The use of a ladder or stepladder is not prohibited by the regulations.

There are many situations where a ladder is the most suitable equipment for working at height.

A decision to use ladders must be justified by the findings of a risk assessment that clearly shows, given all of the circumstances, it is reasonable to use such equipment rather than safer types of access equipment and that any such work is short duration and low risk.



For further information refer to the HSE guidance document *Safe use of ladders and stepladders – A brief guide (INDG455)*.

Some clients, contractors and principal contractors are explicit in their views by not allowing ladders or stepladders on their sites.

They see this as importing an unnecessary risk where, with a small degree of planning, more suitable and ultimately safer equipment may be used.

The tendency of some contractors and sub-contractors to turn up at site with ladders without justifying why safer alternative equipment is not to be used, is being increasingly discouraged.

These regulations are a further stimulus to architects and designers to design out work at height for the construction and ongoing maintenance of buildings, or make safe working at height easier, to further contribute to safety in the industry.

WORK AT HEIGHT REGULATIONS

01

1.3.2 Interpretation of the regulations (Regulation 2)

Below are the most important definitions from within the regulations.

Access and **egress** includes ascent and descent.

Fragile surface means a surface which would be liable to fail if any reasonable, foreseeable load were to be applied to it. This includes the weight of a person or any work equipment or materials.

Ladder includes any fixed ladder or stepladder.

Personal fall protection system means:

- a fall prevention, work restraint, work positioning, fall arrest or rescue system, other than a system in which the only safeguards are collective safeguards; or
- rope access and positioning systems.

Suitable means suitable in any respect, which it is reasonably foreseeable will affect the safety of any person, where, if the measures required by these regulations were not taken, a person could fall a distance liable to cause personal injury.

Work at height means work in any place, including a place at, below or above ground level, along with access to it and egress from it (except by a fixed staircase in a permanent workplace) where, if the measures required by these regulations were not taken, a person could fall a distance liable to cause personal injury.

Working platform:

- means any platform used as a place of work or as a means of access to, or egress from, a place of work
- includes any scaffold, suspended scaffold, cradle, mobile platform, trestle, gangway, gantry and stairway, which is used as a place of work.



Reasonably practicable

The term *reasonably practicable* is used in the regulations. It has the same meaning as it does in the Health and Safety at Work etc. Act 1974 and elsewhere. Therefore, an employer can look at what the risks in a task involving work at height actually are, and what it would cost in terms of time, trouble and effort to avoid them. If it would cost a great deal of money or use of other resources to avoid a very small risk, then it may not be reasonably practicable to avoid it. However, the fact that falls from height often result in fatal or serious injuries should be a considerable factor when deciding what measures are reasonably practicable.

1.3.3 Application and duty of employer (Regulation 3)

The requirements of the regulations that are imposed:

- on an **employer** apply to work carried out by their employees or by any other person(s) under the employer's control, to the extent of that control
- apply to **self-employed** persons, together with persons under their control, to the extent of that control
- apply to persons who are **not self-employed**, in relation to work being carried out for them and under their control, to the extent of their control.

This last provision relates to clients and customers who are having work carried out for them, and the degree to which this duty applies. This depends upon the degree and extent of control that they have over the person(s) carrying out the work. A major client would obviously have more control over a contractor or sub-contractor than, for example, a domestic customer who is perhaps having a roof retiled or gutters cleaned.

1.3.4 Duties of employees

The regulations place a duty on employees, in that they have specific duties under Regulation 14. These duties are similar, and in addition to, their duties under other acts and regulations (such as Section 7 of the Health and Safety at Work etc. Act 1974). In particular, in these regulations, employees must:

- report any activity or defect relating to work at height that they know is likely to endanger the safety of themselves or that of any other person
- use any work equipment or safety device provided for work at height in accordance with any training and/or instructions that they have received.

Given the legal status of health and safety regulations, failure by an employee to comply with the above requirements is an offence that could, in the appropriate circumstances, be dealt with in a court of law or by way of enforcement action.



Unsafe work at height – many factors could have led to these employees thinking this was an acceptable risk

1.4 Organisation and planning of work (Regulation 4)

It is the responsibility of every employer to ensure that all work at height is properly planned, appropriately supervised and carried out in a safe manner, taking into account weather conditions that could jeopardise the health and safety of employees and others that may be affected by the activity.

Planning must include the selection of appropriate work equipment, and planning for any emergencies or any rescue.

An employer will probably meet most of the requirements of this regulation if they have carried out and properly implemented the control measures identified within a suitable and sufficient risk assessment.



For a work at height flowchart refer to Appendix A.



Falling from height

A site supervisor was fined after a 24-year old temporary employee suffered multiple injuries and was left paralysed after falling 10 m from the roof of a warehouse. The temporary worker fell from an unprotected and fragile roof after the supervisor allowed workers who were removing cement sheets to go onto the roof, ignoring the company's safety policy.

The employer, who had a contract to remove asbestos cement roofs from a number of warehouses at the site, had identified them as fragile. The company had agreed a system of work where its employees would use scissor lifts to remove the roof sheets from the underside. However, when the task became difficult the supervisor did not bring the problem to the attention of the contracts manager, as required by the company procedure. Instead he allowed a change to the system of work, whereby he and other employees went onto the roof itself to carry out some of the work.

The HSE found the company's system of work was safe but by changing it the supervisor had sanctioned an unsafe system of work. The company was unaware of the changed way of working and the supervisor had failed to consult with them.

The supervisor pleaded guilty to breaching Regulation 4(1)(b) of the Work at Height Regulations, by failing to properly supervise work at height and make sure it was carried out safely. He was fined £1,000 and ordered to pay £9,765.88 costs.

The HSE inspector said: 'Those who supervise work at height have a responsibility to ensure that it is carried out in a manner that is safe and which guards against the risk of injury from a fall. Such injuries, if not fatal, may result in a lifelong disability for the injured person'.



For further information on fragile roofs refer to Chapter D02 Working at height.

1.5 Competence (Regulation 5)

Employers must ensure that no person takes part in any work at height activity, including being involved in the organisation, planning and supervision of the work and work equipment, unless they are competent to do so or, if they are being trained, that they are being supervised by a competent person.

The level of supervision is important. The supervisor will need to be able to intervene, physically or by virtue of their authority, if an unsafe situation begins to develop, and rectify the situation or stop the work in progress.

The more inexperienced an employee is in relation to working at height, the greater the level of supervision required.

While there is no definition of *competence* in the regulations it may be taken to mean the following.

- A person who has practical and theoretical knowledge of the appropriate aspects of working at height, together with actual experience of what they are to do, which will enable them to ensure that all necessary planning and assessments have been prepared, and safety precautions taken, so that the work may be carried out safely, or that they may work safely. This person must also be able to understand and recognise their own personal limitations.

If they are involved in inspecting work equipment then *competence* may be taken to mean the following.

- A person who has practical and theoretical knowledge, together with actual experience of what they are to inspect, which will enable them to detect errors, defects, faults or weaknesses that it is the purpose of the inspection to discover and to assess the importance of any such discovery. This person must also know their limitations and must not put themselves or others at risk during the inspection of work equipment.

Training is an element of the competence necessary to work at height safely. Similarly, those who deliver such training must also be competent to do so.



To assist trainers in this respect, British Standards publishes the *Code of Practice for the delivery of training and education for work at height and rescue (BS 8454)*.

WORK AT HEIGHT REGULATIONS

1.6 Avoidance of risks from work at height (Regulation 6)

This is one of the main regulations and specifically requires employers to ensure that risk assessments take account of work carried out at height in accordance with Regulation 3 of the Management of Health and Safety at Work Regulations 1999.



For further information refer to Chapter A05 Risk assessments, method statements and permits to work.

In addition, the regulation requires that: 'every employer shall ensure that work is not carried out at height where it is reasonably practicable to carry out the work safely otherwise than at height'.

This clearly requires the employer to carry out a detailed study or assessment of all the ways in which the work could be carried out. If work can be carried out without the need to work at height, then the employer should do so, provided that it is reasonably practicable.



Eliminating the need to work at height

A good example of this was demonstrated during the construction of an over-bridge. The bridge deck was completed on flat ground before the spoil was excavated from below it. This virtually eliminated the need for working at height and reduced the height at which people had to work when constructing the upper parts of the structure.

Ground-level fabrication is another way of eliminating, or at least reducing, some tasks that have been traditionally carried out at height.

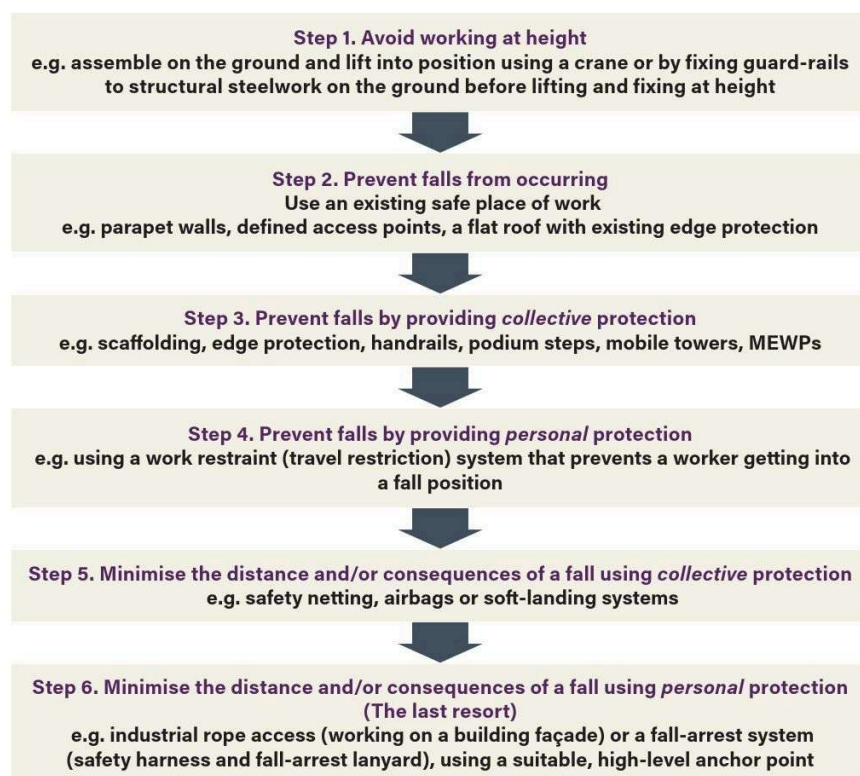
The fact that falls from height often result in fatal or serious injuries should factor heavily in any risk assessment when deciding if an alternative to working at height is reasonably practicable.



When assessing risk or developing a safe system of work, the hierarchy for working at height should be followed.

1.6.1 Hierarchy for working at height

Any person planning work at height should always consider options at the top of the hierarchy before moving down.



Ensure that people with sufficient skills, knowledge, training and experience are selected to perform any task. Any person being trained should be under full supervision at all times.

1.7 Selection of equipment for work at height (Regulation 7)

In selecting work equipment for use in work at height there is a requirement for the employer to consider work equipment or other measures that will give **collective protection** to employees as a whole, rather than just individual personal protection for one person. For example, a guard-rail will protect everyone (collective), whereas a safety harness and lanyard only protects the wearer (personal).

The Work at Height Regulations require that employers follow current industry good practice when selecting equipment for work at height. In particular, employers are required to take account of the following elements.

- Working conditions and risks to the safety of persons at their place of work.
- Access, egress and distances to be negotiated.
- Distance and consequences of any potential fall.
- Duration and frequency of use of the work equipment.
- Need for and ease of evacuation and rescue in any emergency.
- Any additional risks posed by the installation, use or removal of the work equipment, by any evacuation or by any rescue from it.
- Any other provisions of the regulations.

Any work equipment which is selected must meet the following requirements.

- Be appropriate for the work intended to be carried out and the foreseeable loadings.
- Allow the safe passage of persons and materials without risk.
- Be the most suitable equipment for avoiding risks while working at height.

Where a trade or occupation has historically used ladders, stepladders, trestles and other similar work equipment, the regulations do not prohibit their continued use, provided that the provisions of the Work at Height Regulations and the Management of Health and Safety at Work Regulations are fully considered and complied with.

In such circumstances employers will have to be in a position to demonstrate the following.

- The work at height has been properly planned and cannot be avoided or done any other way that is reasonably practicable.
- The most suitable work equipment is proposed to be used, and a risk assessment shows that the remaining risks are low.
- Appropriate steps have been taken to mitigate the effects of any fall, should one occur (for example, safety nets or safety harnesses).
- Employees are competent and properly supervised.

Risk assessments should be thorough but proportionate to the risks inherent in the work that is to be done.

It may, however, be the case that a client or contractor, or principal contractor where appointed, will not allow, or will control through a permit system, certain types of access equipment (such as ladders, stepladders or some types of trestles) on their premises or site.

This requirement may form part of the client's or contractor's own policies and procedures in relation to the use of this form of access equipment. It may mean that an alternative form of access equipment is used.



The HSE has developed a brief step-by-step guide to help workers at height understand the key factors to consider when selecting the safest and most appropriate types of access equipment.

1.8 Requirements for particular work equipment (Regulation 8)

This regulation and Schedules 2 to 6 of the regulations cover the following provisions.

- Guard-rails, toe-boards, barriers and similar means of protection.
- Working platforms (such as scaffolds and trestles).
- Safety nets, airbags and other collective means of arresting falls.
- Personal fall protection systems.
- Work positioning systems.
- Rope access and positioning systems.
- Work restraint systems.
- Ladders.
- Fall-arrest systems.

In respect of guard-rails, the main guard-rail must be installed to at least 950 mm above the work surface or the edge from which a person may fall.

WORK AT HEIGHT REGULATIONS

01

Although there is no statutory minimum height for toe-boards, both BS EN 12811-1 and the current industry guidance, TG20:21, which relate to scaffolding good practice, recommend a minimum of 150 mm. They must be suitable and sufficient to prevent the fall of persons or materials. In line with current industry practice it is anticipated that scaffold boards on edges will continue to be used in most circumstances.

The maximum distance or gap between the top of a toe-board and the mid guard-rail, or between the mid guard-rail and the top guard-rail, is 470 mm.

Where brick guards or similar items are used as a means of protection, then they must meet the following criteria.

- Be placed so as to prevent the fall of persons or materials.
- Be of a suitable size and strength.
- Be placed or secured so that they do not become accidentally displaced.
- Be inspected at regular intervals.

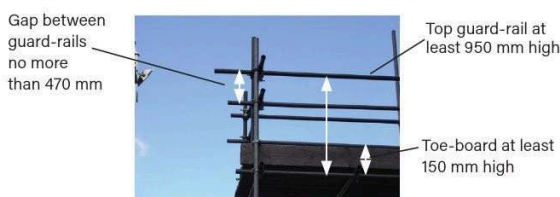
Guard-rails may be removed on a temporary basis for the movement of materials, provided that suitable and effective alternative fall protection measures are put in place for the duration of the work, and that the guard-rails are replaced as soon as possible after the work is completed.

There is also provision in the regulations for danger areas (*refer to Regulation 11*). Where a workplace contains an area where there is a risk of injury to any person at work falling, or being struck by a falling object, measures should be taken to prevent unauthorised access and the danger area must be clearly indicated. This would generally mean the provision of an exclusion zone, supported by safety signage. The safety signage must comply with the Health and Safety (Safety Signs and Signals) Regulations 1996.

Where employers can demonstrate that they are complying fully with current industry good practice and HSE Codes of Practice or guidance, then in the majority of situations they will meet most of the requirements of the Work at Height Regulations. Their risk assessments for working at height will then essentially be a comparison between what they are already doing to control the risks and what is regarded as good practice, then implementing further control measures if necessary.



Members of the public should be protected from items falling from height



Guard-rail and toe-board dimensions

1.8.1 Use of scaffolds

The current industry guidance is based on the content of the British Standard: *Temporary work equipment. Part 1. Scaffold – Performance requirements and general design* (BS EN 12811-1) and *The standards for temporary edge protection systems* (BS EN 13374).

Additional practical guidance is produced by the National Access and Scaffolding Confederation, which produces the *Guide to good practice for scaffolding with tubes and fittings* (TG20) and *Preventing falls in scaffolding* (SG4). Both documents are accepted by the scaffolding industry and the HSE as essential in ensuring that scaffolding is designed and erected in a safe manner, and in compliance with the Work at Height Regulations. Industry trade bodies, including the Scaffolding Association and system scaffold manufacturers, also publish advice, technical information and guidance for scaffolders and their clients, on industry standards, scaffold design and health and safety in scaffolding operations.

1.8.2 Use of ladders, stepladders and trestles

Ladders may be used for access to and egress from the place of work. Ladders, stepladders and trestles may be used to work from, provided that all of the requirements of the Work at Height Regulations are fully complied with.

The primary objective of the regulations is to eliminate work at height (and so the possibility of falls) and, where this is not possible, to ensure that all work at height is carried out safely and that the work equipment being used (such as a ladder or stepladder) is appropriate, suitable and safe for the work and the environment in which they are to be used.

In deciding to use a ladder for access or egress, or a ladder, stepladder or trestle for work at height, the employer must have carried out an assessment and found that it is not reasonably practicable to use any safer means, and that a risk assessment has shown that the risks from using the ladder, stepladder or trestle are low.

Ladders should be used in accordance with manufacturers' instructions. They must be the right ladder for the job, positioned at the correct angle, placed on a firm, level surface, and secured or otherwise prevented from slipping or moving. Users should face the ladder at all times when climbing up or down, have both hands free, and not be carrying anything that would interfere with their safety or balance.

Ladders may be used as a place of work, subject to the above, and if it is light work and of short duration (working on a ladder for no more than 30 minutes at a time). Users should be trained in how to safely erect and safely work on ladders (for example, not to overreach).

When stepladders are to be used to carry out work they should be of a suitable size and type. Generally, stepladders should be placed facing the work. They should not be used side-on to the work in any situation where a sideways load could be applied. Again, the work should be light and of short duration.

Pre-fabricated access equipment (such as podium steps) are a product that was developed as a result of the implementation of the Work at Height Regulations and, in simple terms, are a stepladder with a secure working platform with guard-rails and toe-boards in place. They are available in various sizes and heights and many are fitted with castors to make them more mobile. This type of access equipment provides a safe working platform and collective means of fall protection.

Trestles should be of sound construction, and properly set up on a surface that will bear their weight, as well as any loading of persons or materials. Trestle boards must be supported at a distance specified by the manufacturer. If they are adjustable, using telescopic adjustment, they should have high tensile pins in the adjusters. If they are fitted with wheels or castors, they should have brakes or other suitable locking devices on the wheels or castors. Employees who install and use trestles should be trained how to do so safely.

Trestles should be fitted with guard-rails and toe-boards, where appropriate.



The Ladder Association, in consultation with the HSE, has produced risk assessment guides for ladders and stepladders.



For further information on the use of ladders refer to Chapter D03 Common access equipment.

1.9 Fragile surfaces (Regulation 9)

The general requirement is that employers must ensure that no person passes across or near, or works on, from or near any fragile surface when it is reasonably practicable to carry out the work safely without their having to do so.

Where this requirement cannot be met, then the employer must:

- provide and ensure that there are suitable and sufficient platforms, coverings, guard-rails or other similar means of support or protection, which must be capable of supporting any foreseeable load or loading
- where the risk of a fall still remains, take suitable and sufficient steps to minimise the distance and consequences of any fall, should it occur
- place prominent warning notices at the approach(es) to any fragile material
- where such notices cannot be used, ensure that employees (and others as appropriate) are made aware of the fragile materials by other means.

1.10 Falling objects (Regulation 10)

The requirement in this regulation follows current good practice in that employers must take suitable and sufficient steps (including prohibiting the throwing down of materials) to prevent, so far as is reasonably practicable, the fall of any materials or objects that are likely to cause injury to any person. Suitable steps will include the use of brick guards, toe-boards and debris nets.

Where there is a residual likelihood of falling objects, or in the interests of safety, suitable and sufficient measures must be taken to prevent persons from being hit by falling objects or materials. This will cover the use of protective fans, boarded or roofed walkways and exclusion zones at ground level, which must all be supported with the appropriate level of safety signage.

Materials on scaffolds and working platforms must be stored so that they cannot fall or pose a risk of injury to anyone by their collapse, overturning or unintended movement.

1.11 Danger areas (Regulation 11)

Following from Regulation 10 and building on its requirements, in any workplace where there is a risk of any person falling or of persons being struck by falling objects, employers must take all reasonably practicable steps to prevent any unauthorised access to that area (such as putting up physical barriers and appropriate warning signs). Any signs must comply with the Health and Safety (Safety Signs and Signals) Regulations 1996.

1.12 Inspection of work equipment (Regulation 12)

This regulation only applies to work equipment to which Regulation 8 and Schedules 2–6 apply, and closely follows the current requirements in the Provision and Use of Work Equipment Regulations 1998 as regards the inspection of work equipment.

Where the safety of equipment used for working at height (for example, a scaffold) depends upon how it has been installed or assembled, then it must be inspected by a competent person before being used.



Throughout the regulations, *inspection* means any visual or more rigorous inspection, and any appropriate testing that a competent person decides is necessary.

All work equipment exposed to conditions causing deterioration that may result in dangerous situations must be inspected. This will ensure that it remains safe and that any deterioration can be detected and remedied.

WORK AT HEIGHT REGULATIONS

01

Inspections must take place at suitable intervals and after exceptional circumstances have occurred that are liable to have jeopardised the safety of the work equipment.

In addition to these requirements, any working platform that is used for construction work, and from which a person could fall more than 2 m, must not be used unless it has been inspected in that position within the previous **seven days**. A mobile working platform (such as a mobile scaffold tower or a MEWP) must have been inspected within the previous seven days.

Employers must ensure that work equipment (other than lifting equipment for which a similar provision applies under the Lifting Operations and Lifting Equipment Regulations 1998 (LOLER)), whether it is purchased, brought in or hired, is not used unless they have evidence that the inspection records are compliant with regulations and up to date (*see 1.14.7 below*).

If there is no evidence of the last inspection then the work equipment cannot be used on site until an inspection is carried out by a competent person. Formal inspection should not be regarded as a substitute for any routine pre-use checks that should be carried out by the user.

Inspection intervals should be decided by a competent person, based on the results of risk assessments, and then be reviewed in the light of experience.

The purpose of an inspection is to identify whether the work equipment is safe to use, and that any defect or deterioration is detected and repaired. If this is not possible, the work equipment should be removed from service before it becomes an unacceptable risk.

The results of every inspection must be recorded and kept on site until the construction work is completed, and after that they must be kept at one of the company offices for three months. The detail of the inspection must be recorded before the end of the working day, in accordance with the particulars given in Schedule 7, and the report delivered within 24 hours.

Lifting equipment that has had a thorough examination under the provisions of LOLER is treated as having had an inspection under these regulations.

Provided that the required information is recorded, and is readily available, then an employer may adopt their own recording system. Any computer-based system should be secure and cannot be interfered with. Systems may also be in the form of a register.

1.13 Inspection of places of work at height (Regulation 13)

So far as is reasonably practicable, every employer must ensure that the surface and every parapet, permanent rail or other such fall protection measure be inspected visually prior to each use. While there is no requirement to record such inspections, a simple record (a note in the site diary for example) would provide evidence that the risk has been identified and appropriate inspections carried out.

1.14 Schedules 1 to 8

There are eight schedules to the regulations. Much of the information contained in the schedules is virtually identical to current requirements and/or guidance. Therefore, if employers are actively working to current standards and guidance then there is little extra for them to do.

1.14.1 Schedule 1. Regulation 6(4)(a)

The requirement in this schedule is for existing places of work at height and means of access to or egress from such places to be stable, of sufficient size and strength and with suitable edge protection as necessary. Ladders, in particular, must rest on stable, strong surfaces.

A place of work is to be properly constructed, used and maintained so as to prevent the risk of tripping, slipping or being trapped between it and adjacent surfaces. It should have no gaps through which materials could fall and injure someone below.

1.14.2 Schedule 2. Regulation 8(a)

This covers the requirements for guard-rails, toe-boards, barriers or similar means of collective protection (*refer to Regulation 8*). The remainder of the requirements are unchanged, other than the fact that the top guard-rail must now be at least 950 mm above the edge from which any person might fall (with the provision of 910 mm for fixed guard-rails that were in place at the time the Work at Height Regulations came into force), and the removal of any specific height above ground level at which guard-rails and toe-boards must be fitted.

1.14.3 Schedule 3. Regulation 8(b)

This covers the requirements for working platforms. Part 1 deals with requirements for all working platforms, and Part 2 covers the additional requirements for scaffolding.

Part 1. All working platforms must be erected and used so that components cannot become displaced and cause danger to anyone. They must be suitable for the work, and of sufficient strength and rigidity. If they are altered or modified (by a competent person) they must remain safe and stable.

A **supporting structure** means any structure used for the purpose of supporting a working platform and includes any plant used for that purpose. It must be suitable and of sufficient strength and stability while being erected, used or dismantled. It must be prevented from slipping or moving, and if it has wheels or castors, they must be capable of being locked or similar. Working platforms and supporting structures must not be overloaded.

Part 2. As regards the **additional requirements for scaffolding** strength and stability, calculations must be carried out unless it is being assembled in conformity with generally recognised standard configurations (for example, a TG20-compliant scaffold, erected in accordance with a TG20 compliance sheet).

If the scaffold is not a TG20-compliant scaffold then design calculations will need to be provided by the scaffolding contractor. Such calculations are usually supported by the provision of detailed drawings, to assist scaffolders during construction and inspection of the scaffold.

If prefabricated scaffolding materials are being used (for example, ladder beams and hop-up platforms), then the manufacturer's data will be required to confirm that these products may be used in conjunction with the scaffold in question.

Depending on the complexity of the scaffold, an assembly, use and dismantling plan (commonly referred to as a method statement) shall be drawn up by a competent person. This may be a standard or generic plan, supplemented with specific details, as appropriate.

A copy of the plan must be kept for the use of any persons concerned with the erection, use or dismantling of the scaffold.

The size and layout of the scaffold must be appropriate and suitable for the work to be performed, and it must also permit the work and passage of persons to happen safely. The detailed design drawings or TG20 compliance sheet for the scaffold will give details on specific scaffold dimensions, layouts and load bearing capacities.

When a scaffold is not available for use during erection, alteration and/or dismantlement, then warning signs in accordance with the Health and Safety (Safety Signs and Signals) Regulations need to be displayed (scaffold tagging systems are available that will help fulfil this requirement).

Physical barriers must also be put in place to prevent access to the scaffold, or any unsafe or incomplete parts of the scaffold.



Scaffolding may only be assembled, dismantled or altered under the supervision of a competent person and by persons who have received appropriate and specific training.



The Construction Industry Scaffolders Record Scheme (CISRS) has a card scheme in place for people working in the scaffolding industry, including scaffolders, trainee scaffolders, labourers, supervisors and inspectors.

1.14.4 Schedule 4. Regulation 8(c)

This schedule covers the requirements for collective safeguards for arresting falls. Collective safeguards include safety nets, airbags, landing mats and any similar devices or arrangements. All must be suitable and of sufficient strength to be able to safely arrest or cushion a fall and may only be used in the following situations.

- Where a risk assessment has shown that the work to be done can be carried out safely while the safeguard is being used, and without influencing its effectiveness.
- Where the use of safer work equipment is not reasonably practicable.
- If a sufficient number of employees (or others) have received adequate training specific to the safeguard, including rescue procedures.

An important requirement is that if the safeguard is designed to be attached to a building or a structure then the safeguard, the structure and all anchorages must be suitable and of sufficient strength for any foreseeable load that a fall might impose on it.

Airbags and landing mats must be suitable for the purpose intended, and they must be stable.

If a safeguard is designed to distort when arresting a fall, it must give sufficient clearance from the ground or adjacent structures to avoid injury to a person whose fall is being arrested.

1.14.5 Schedule 5. Regulation 8(d)

This schedule covers the requirements for personal fall protection systems (Part 1), work positioning systems (Part 2), rope access and positioning techniques (Part 3), fall-arrest systems (Part 4) and work restraint systems (Part 5).



Much of the information in this schedule is technical in nature and it is of paramount importance that employees are fully trained in the use of any of the fall protection systems available to them. There is insufficient detail within this brief explanation of the schedule to train a person on the requirements, or for them to gain sufficient knowledge to be regarded as competent.

All safety harnesses, lanyards and other fall prevention or fall-arrest equipment must comply with the appropriate British and European standards.

Part 1 deals with the requirements for all personal fall protection systems. See 'Interpretation of the regulations' (refer to 1.3.2) for the definition of personal fall protection systems.

Current guidance is that the requirements set out in this schedule apply to all rope-based activities for work at height, including industrial rope systems and any other similar activity when carried out as a work activity.

WORK AT HEIGHT REGULATIONS

01

The following points are requirements for personal fall protection systems within the schedule.

- Shall only be used if a risk assessment has shown that the work can be done safely while it is being used, and that the use of other, safer work equipment is not reasonably practicable.
- The user and a sufficient number of others have been trained in its use and in rescue procedures.
- It shall be suitable and of sufficient strength for the purpose for which it is to be used and will withstand any foreseeable loading.
- It fits the wearer and is correctly fitted or worn.
- It is designed to minimise injury to the user in the event of a fall, and is such that the user will not fall or slip out of it should they fall.
- It is designed, installed and used so as to prevent unplanned or uncontrolled movement of the user.

Any anchorage point must be suitable and of sufficient strength to support any foreseeable loading. If designed to do so, the equipment must be securely attached to at least one such anchorage point when in use.

Part 2 deals with the additional requirements for work positioning systems. They may only be used in the following instances.

- If the system includes a suitable backup system for preventing or arresting a fall.
- If the system includes a line as a backup system and the user is connected to it.
- Where it is not reasonably practicable to do either of the above, then other suitable measures are taken to prevent or arrest a fall.

Part 3 deals with the additional requirements for rope access and positioning techniques, which may only be used in the following instances.

- If it has two separate lines: the working line and a safety line.
- If the user has a safety harness that is connected to both the working line and the safety line.
- When the working line has safe means of ascent and descent, and a self-locking device to prevent falling.
- When the safety line has a mobile fall protection system that is connected to, and travels with, the user.
- Subject to the type and duration of work, and the findings of a risk assessment, it has a seat with appropriate accessories. However, if a risk assessment has demonstrated that a second line would entail a higher risk to the user, then, provided that appropriate safety measures have been taken, a single rope may be used.

Part 4 deals with the additional requirements for fall-arrest systems. It requires that they must have a suitable energy absorber (often called a shock absorber and which is usually either a folded metal strip that deforms, or tear-away stitched webbing), or other suitable means of limiting the force applied to the user's body in the event of a fall. In addition, a fall-arrest system must not be used if there is any risk of a line (a rope or a lanyard) being cut, or where there is no safety zone or clear zone to allow for any swinging or pendulum effect after a fall, or in a way that hinders the system's safety performance or makes its use unsafe.

Part 5 deals with the additional requirements for work restraint systems (often a safety harness with a very short lanyard), and requires that they are designed so that they are used correctly to prevent the user from getting into a position where they could fall.

1.14.6 Schedule 6. Regulation 8(e)

This schedule contains a specific requirement on the use of ladders and stepladders for work at height. Employers and the self-employed must ensure that a ladder or stepladder is only used for work at height if a risk assessment has shown that the use of more suitable work equipment is not justified because of the low risk and the short duration of the work, or because of existing features on site that cannot be altered.

The remainder of the schedule is in line with current industry good practice. The requirements are shown in brief below.

- The surface on which a ladder rests must be stable and of sufficient strength.
- The ladder is strong enough for loads that may be put on it.
- The ladder is placed so that it is stable during use.
- A suspended ladder is attached in a secure manner so that it does not swing.
- Portable ladders are prevented from slipping by being secured at or near their top or bottom, or with anti-slip or stability devices, or other effective means.
- Access ladders are long enough to provide a handhold when getting off at the top, unless other suitable handholds have been provided.
- Sections on interlocking or extension ladders are prevented from moving while in use.
- Mobile ladders are prevented from moving before being stepped on.
- Where reasonably practicable, rest platforms are provided where a run of ladders rises a vertical distance of more than 9 m.
- Ladders are used in such a way that a secure handhold and secure support are always available to the user.
- The user can maintain a safe handhold while carrying a load.

This last point is qualified for stepladders in that:



... unless, in the case of a stepladder, the maintenance of a handhold is not practical when a load is carried, and a risk assessment has demonstrated that the use of a stepladder is justified because of the low risk and the short duration of the work.



For advice on footing ladders, refer to Chapter D03 Common access equipment.

1.14.7 Schedule 7. Regulation 12(7)

This schedule states the details to be recorded in a work equipment inspection report (for example, scaffold inspection), as follows.

- The name and address of the person for whom the inspection was carried out.
- The location of the work equipment inspected.
- A description of the work equipment inspected.
- The date and time of the inspection.
- Details of any matter identified that could give rise to a risk to the health or safety of any person.
- Details of any action taken as a result of any matter identified in the point above.
- Details of any further actions considered necessary.
- The name and position of the person making the report.

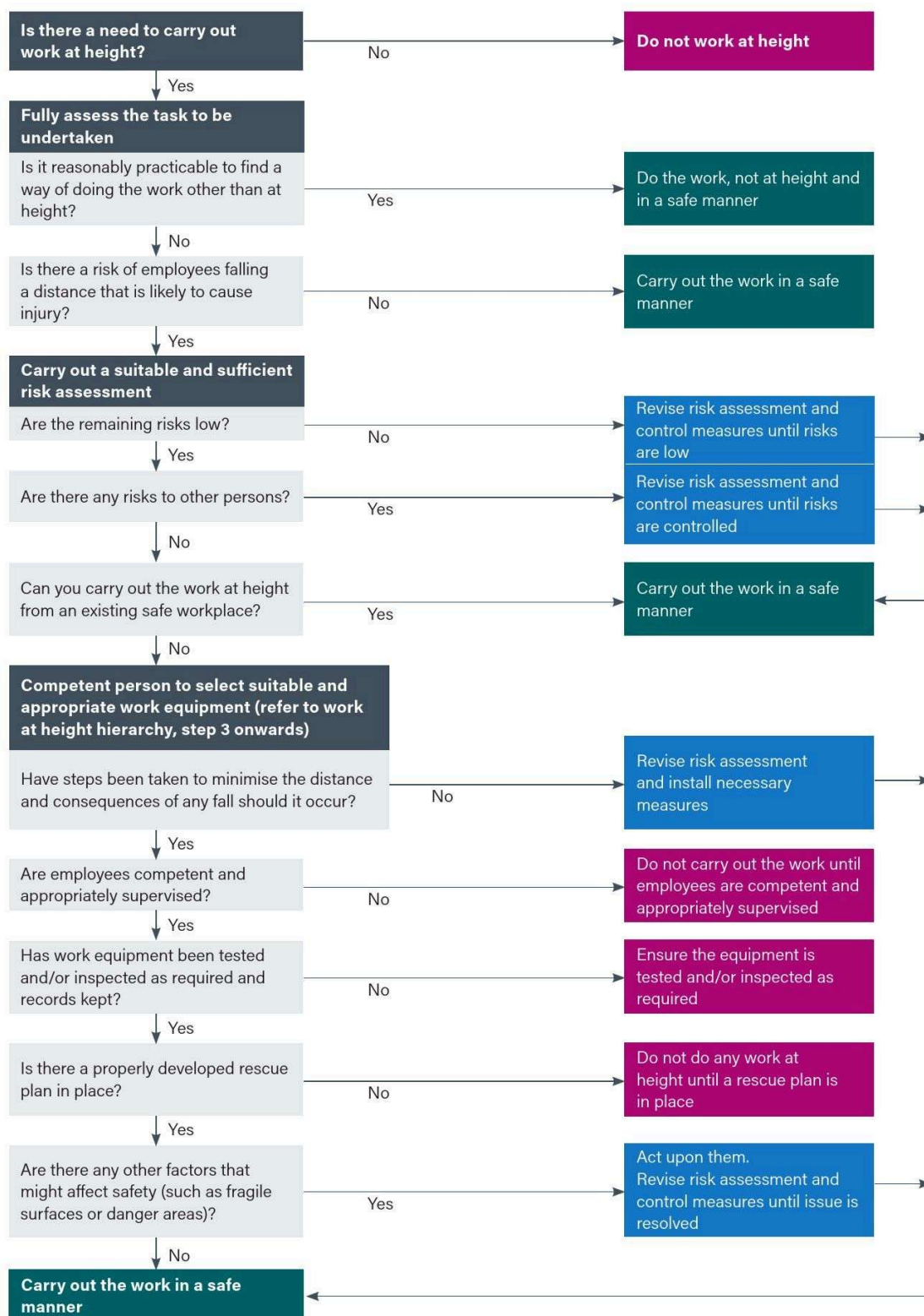
Reports of inspection have to be kept on site until construction work is completed and then at the employer's offices for three months.

1.14.8 Schedule 8. Regulation 19

This schedule outlines various other regulations or parts of them that are revoked by the coming into force of the Work at Height Regulations.

WORK AT HEIGHT REGULATIONS - APPENDIX A

Appendix A - Work at height flowchart



CONTENTS

Working at height

02

2.1	Introduction	16
2.2	Important points	17
2.3	Legislative requirements	17
2.4	Stages of a roofing project	19
2.5	Types of roof work	22
2.6	Fragile roofs and surfaces	25
2.7	Managing asbestos in roof work	27
2.8	Access arrangements	27
2.9	Handling and storage of roofing materials	34
2.10	Other considerations	35

Supporting INFORMATION

GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.



**Scan the QR code to complete a brief survey,
and share your feedback on the course.**

Overview

There is no such thing as a safe height; anyone who is working at height is at risk of falling. This hazard exists on working platforms, scaffolds, ladders, flat and pitched roofs, open steelwork and any area where work is carried out near fragile materials, openings, holes and roof edges.

Ladders, stepladders and trestles (or lightweight staging) are among the most commonly used pieces of access equipment on site and, perhaps, the most misused. Most construction workers have used them at some point and it is essential that safe working practices should be followed if accidents are to be avoided.

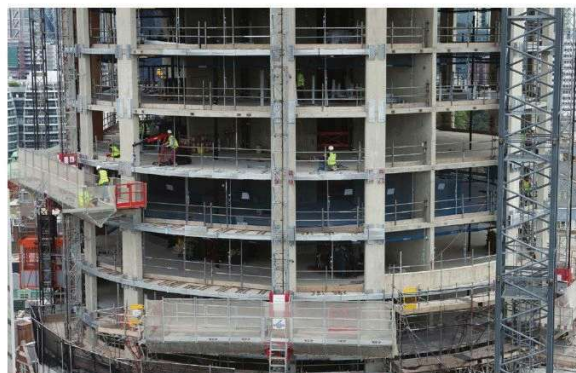
2.1 Introduction

Before you read this chapter, it is strongly recommended that you familiarise yourself with Chapter D01 Work at Height Regulations 2005. There have been a number of major improvements in safety standards and practices with regard to working at height. For example, the standard practice of profile roofing workers using safety nets on new-build projects and steel workers using powered access have been significant steps forward in construction health and safety.

Every year, however, falls from height remain by far the largest cause of death in the construction industry. Roof work in particular, including work on fragile roofing materials, gives rise to a substantial number of fatal and serious accidents. These invariably affect lives and livelihoods.

It needs to be emphasised that there is not just the actual roof working to be considered but the whole process: planning the job, creating a safe access, the safe storage of equipment and materials, and so on.

In many respects, working for domestic clients on maintenance or repair work may offer greater challenges to providing a safe working environment. The work is typically of shorter duration than new build, and often carried out directly for a client who may not be aware of the required standards. This does not lessen the need for safe working methods and compliance with the law for those undertaking the work. Compliance with legislation and well-established codes of practice could save many lives and prevent many injuries. Most accidents could be avoided with provision of appropriate equipment and the adequate information, instruction, training and supervision of those who use it.



Various work at height solutions in practice, including proprietary edge protection mesh barriers and scaffold guard-rails

2.1.1 Planning, design, construction and maintenance of tall buildings

With the number of tall buildings within the United Kingdom, particularly in London, increasing rapidly, a working group was put together by the Construction Industry Advisory Committee (CONIAC) consisting of the Access Industry Forum, members of the construction industry, and those associated with it, all of whom have knowledge and experience of the planning, design, construction and maintenance of tall buildings. The group has worked together to produce a guidance book called 'Industry guidance for the planning, design, construction and maintenance of tall buildings'. This is freely available, and provides guidance on how to follow a 'risk management approach' to identify hazards and provide solutions to enable the design and safe construction of tall buildings.



For further information and a link to download the guidance, please visit the Construction Industry Publications website.

Following the Grenfell Tower tragedy, a new regulatory framework was established for duty holders involved with the planning, design, construction and refurbishment of a building to ensure safety across the building's life cycle. The Building Safety Act 2022 sets out safety requirements for landlords of higher-risk buildings (at least 18 m or seven storeys high, with two or more residential units). The Act has six parts, and contains provisions intended to secure the safety of people in or about buildings as well as improve building standards.



For further information on the Building Safety Act 2022, visit the Health and Safety Executive's (HSE's) website.

For additional information refer to:



- Chapter D01 Work at Height Regulations
- Chapter D03 Common access equipment
- Chapter D04 Scaffolding
- Chapter D05 Fall arrest and suspension equipment
- Chapter F05 Working over or near to water.

For further sources of information and guidance relating to roof work visit the following websites.



- **Advisory Committee for Roofsafety (ACR).**
- **National Federation of Roofing Contractors (NFRC).**
- **Access Industry Forum (AIF).**
- **National Access and Scaffolding Confederation (NASC).**
- **HSE.**
- **British Constructional Steelwork Association (BCSA).**

2.2 Important points

- Those who work on roofs are, by the nature of their work, at risk of falling if appropriate measures are not taken.
- Those who plan, supervise or carry out roof work must be competent to do so.
- A risk assessment and a method statement should be compiled for all such work.
- Integral features of roofs (such as a steep pitch, valleys, fragile cladding or roof lights) can increase the risk of falling.
- Falls through fragile roofing materials continue to be the cause of many deaths and injuries. Where such surfaces have been identified, proprietary covers (either temporary or permanent installations) prevent those passing or working nearby from falling through.
- Bad weather can have a significant impact on the risk control measures that have to be taken.
- Ideally, work will be carried out from a stable working platform, fitted with guard-rails and toe-boards, all of which comply with the requirements of the Work at Height Regulations.
- If safe working platforms cannot be provided, then safety harnesses with work restraint systems should be used to prevent falls.
- Where safety harnesses with work restraint systems cannot be used, effective fall-arrest measures must be put in place, ideally collective measures (such as safety nets or other soft-landing systems).
- People who work on roofs must consider the safety of anyone passing below to protect them from falling materials or tools.
- Access equipment (such as powered access (MEWPs), scaffolding, ladders, stepladders and lightweight stagings) are all easily transported means of access to work at height. They can all be used safely in certain conditions.
- Danger often occurs when access equipment is used for a job for which it is not suitable; this particularly applies to ladders and stepladders.
- All persons working at height must have received adequate training in systems of work and any equipment that they are planning to use for access.
- The nature and duration of the job will be significant factors in determining the most appropriate type of access equipment to use for work at height.
- The HSE's evaluation of accidents resulting from work at height shows that inappropriate access equipment, provided by management and used by workers, is a significant factor.
- Falls from low levels (such as a height below 2 m) are the cause of many injuries and fatalities.
- All openings and holes in floors must be protected with secured guard-rails, toe-boards, coverings or similar barriers and be clearly marked with warning signs.

2.3 Legislative requirements

2.3.1 Work at Height Regulations 2005

The relevant requirements of these regulations with regard to roof work are considered to be as follows.

- Employers should ensure that work at height is planned, supervised and carried out in a safe manner by trained and competent persons.
- Work at height must be carried out using appropriate work equipment, particularly that which provides collective fall protection.
- Suitable and sufficient steps should be taken to prevent falling objects that are likely to cause injury to any person.
- Consideration needs to be given to the weather conditions.
- Plan for the emergency rescue of employees from height and where appropriate make specific arrangements for how to rescue someone who has fallen but is suspended.
- The method of rescue needs to be proportionate to the risk and should not rely on the emergency services.
- Where there is a risk of a person falling or being struck by a falling object, steps need to be taken to prevent unauthorised access into that area.

WORKING AT HEIGHT

There are a number of trade associations for roofing companies (listed below) that could be consulted on, or that comment upon, queries about competence.

- National Federation of Roofing Contractors (NFRC).
- Flat Roofing Alliance (FRA).
- Single Ply Roofing Association (SPRA).
- Mastic Asphalt Council (MAC).
- Rural and Industrial Design and Building Association (RIDBA).

Whilst membership of these organisations by itself does not guarantee either good health and safety practices or quality, they do all vet members and operate a dispute resolution service, which may be a useful facility.



The Advisory Committee for Roofsafety is a body dedicated to making working on roofs safer. It was established in 1998 and is made up of nominees from HSE, trade associations and organisations involved in roof work that provide many years of experience of involvement in working on roofs.

Visit the Access Industry Forum website for some useful online videos.

2.3.2 Construction (Design and Management) Regulations 2015 (CDM)

The regulations place legal duties on duty holders, each of which has the potential to reduce the risks to health and safety during and after the construction phase.



For additional information and resources visit the HSE website.

2.3.2.1 Designers

The definition of **designer** under CDM is broad. It can include architects, consulting engineers, quantity surveyors, interior designers, temporary work engineers, chartered surveyors, technicians or anyone who specifies or alters a design. Therefore, principal contractors, specialist contractors and individual tradespersons could also be designers. A common scenario would be where the client asks the contractor to sort out a leaking roof and the contractor designs the solution. A commercial client who becomes actively involved in designing their project may also be considered to be a designer.

Whoever prepares or modifies a design for a construction project (including temporary works) or instructs someone else to do so will be considered to be a designer. This is very important as all designers have a legal duty to consider health and safety issues in relation to not only carrying out the work, but also the maintenance, cleaning and eventual removal (demolition) of the roof.

In common with all aspects of design, the person carrying out the design should be sufficiently knowledgeable of the construction process to specify how the work can be carried out safely.

Roofers working on industrial type buildings are commonly faced with the problem of fragility. This may be because the roof itself is made of a fragile material (such as asbestos cement sheets) or because the roof lights are not load bearing. As the standard specification for roof lights is 10% by area, this represents a significant amount of fragile roof surface.

The HSE states quite categorically that stringent effort should be made by designers to ensure that fragile roof lights are not specified.



The Advisory Committee for Roofsafety (ACR), which is the lead authority in the UK on roof work health and safety and represents all the relevant trade associations, has published jointly with the HSE definitive guidance on what constitutes fragile material: *Red book - Test for non-fragility of large element roofing assemblies* (ACR[M] 001:2019).

This book details tests that manufacturers should carry out on profiled roof sheets. Anyone who specifies this type of roofing must be aware of the contents of this book and ensure the materials they specify meet the standards, where possible.

When designing roofs, designers should also consider such things as ongoing maintenance work (for example, how roof lights and gutters can be cleaned safely) and whether the safe access for this type of work can be designed-in at the design stage.

2.3.2.2 Clients

Under the regulations the client has the main duty for providing relevant pre-construction information to each designer (including the principal designer) and contractor (including the principal contractor) being considered for the job, or already appointed to it, to enable the job to be planned so that it can be carried out safely. For projects involving more than one contractor the client can expect help from the appointed principal designer.

The main type of information that would be relevant for roof work would be the presence of asbestos, areas of damage, loading limitations of the roofing material, the existence of fall-restraint systems, the location of safe access routes where known, as-built drawings, where available, and so on.

It should be noted that the client may not be aware of such factors as damaged areas of the roof or the existence (or lack) of safe access routes. The client may assume that the job will be carried out by a method that is neither practical nor safe. They commonly expect roofing contractors to be able to access places and do tasks that they would simply not allow their employees to do.

2.4 Stages of a roofing project

The various stages of a roofing project are shown below.

- Design.
- Selection of contractors and staff.
- Planning.
- Carrying out the work.
- Post-completion information (at the end).

2.4.1 Stage 1. Design

Designers have duties under CDM to eliminate, reduce or control foreseeable health and safety risks that may arise during construction, use and future maintenance of roof structures. Where residual hazards remain in the design, these should be communicated in the planning so that suitable precautions may be considered.

A designer must take all reasonable steps to provide, with the design, sufficient information about the design, construction or maintenance of the structure, to adequately assist the client, other designers and contractors to comply with their duties under these regulations. This information will assist all duty holders, during design, construction and for the lifetime of the structure.

Designers should take into account the specification of materials and construction methods, including fragility. Particular consideration should be given to future requirements for maintenance, cleaning and access to the finished project. Discussing these requirements with those who will be carrying out this work is important. They may have established methods of working, or specific needs or suggestions, which will need to be considered in the design.



Design includes drawings, design details, specifications and bills of quantities (including specification of articles or substances) relating to a structure, and calculations prepared for the purpose of a design.



For further information refer to Chapter A03 Construction (Design and Management) Regulations.

2.4.2 Stage 2. Selection of contractors and staff

Where a company is using its own employees to carry out roof work, it is essential that several factors are taken into consideration. Operational work on roofs can be both hazardous and strenuous, often involving:

- work at considerable heights for long periods of time
- work outdoors, including adverse weather conditions (such as hot, cold or wet weather, high winds, snow, ice and fog)
- repetitive materials handling
- reaching, stretching and maintaining balance in awkward postures whilst carrying loads on varying roof terrains.

This means that fitness to work is particularly important and needs to be considered by the employers. The Advisory Committee for Roofsafety (ACR) suggests that it is vital that people working on roofs do not suffer from the following conditions.

- Any neurological condition likely to cause a seizure.
- Weakness of limbs.
- Loss of balance, including vertigo (dizziness from being at height).
- Any heart or lung condition likely to be aggravated by strenuous work.
- Any disability or impairment of limb function.
- Any other disease, disability or the effects of medication, alcohol, drugs or toxic substances (lead and so on) likely to impair mental or physical activity, especially at a height.
- Temporary ailments (such as influenza) or other conditions that may affect judgement.
- Uncorrected sight problems.
- A physique that would be unsuitable for the work.

Obviously an employer needs to be aware of the provisions of the Equality Act 2010 with regard to the above. It is strongly suggested that employers should establish a policy on fitness for work, and ideally that this should be implemented before taking on any new staff.

So-called fitness to work medicals are not new and many sectors already use them for their staff. Often a local GP or a specialist occupational health service provider can carry out these medicals.



For further information refer to the ACR's *Black Book*; Guidance note for competence and general fitness requirements to work on roofs.

WORKING AT HEIGHT

2.4.3 Stage 3. Planning

Planning the work should include consideration of the progression of the work with regard to the following.

- Site-specific hazards and risks.
- Weather conditions.
- Emergencies (including rescue).
- Safe means of access and egress.
- Materials handling and storage.
- Personnel involved, including competence and supervision.

2.4.3.1 Site-specific hazards and risks

Site-specific hazards could include the following.

- Working above public areas (such as shopping malls or public streets).
- The difficulties in delivering materials and transferring them to roof level.
- The presence of site traffic or road traffic on a public road.
- Awkward working environments (such as occupied houses or factories).
- Nesting birds that will aggressively defend their territory.
- The presence of accumulated pigeon droppings (*refer to Chapter B01 Management of health*).
- Emergency situations (such as rescuing someone who has fallen and is suspended at high level in a safety harness or recovering an injured worker from height).
- Vent pipes that may suddenly shower unsuspecting roof workers with anything ranging from high pressure steam to noxious chemicals.

2.4.3.2 Weather conditions

The weather can have a significant impact upon the intention to carry out roof work and may ultimately be the reason for the start of a job being delayed, or suspended part-way through. Heavy rain, high winds, frost, snow or fog might make it unsafe to work at height.

! It is essential to monitor wind speeds on a regular basis at the relevant working height.

Wind speeds and temperatures quoted in weather forecasts can differ considerably at height (for example, if installing a glass atrium roof on top of a multistorey tower block). If a roofing job has started and the weather is forecast to be changeable, with perhaps extremes of conditions, it will be essential to monitor the forecast and conditions so that work can be halted before it becomes unsafe to continue.

Wind speeds in excess of 25 mph can create unsafe working conditions. Winds can funnel and eddy around buildings, causing turbulence, which again may make working places unsafe. In cold weather conditions, the wind will effectively lower the temperature considerably – this is known as the wind chill factor.

Work involving the handling of sheeting and cladding requires extra care in windy conditions, when a sheet may act like a sail, causing the person holding it to lose their balance. Guidance from the National Federation of Roofing Contractors (NFRC) suggests that the following work should cease when the average wind speeds shown are exceeded.

17 mph – handling lightweight materials and any materials over 5 m long or rolls of felt.

23 mph – general roofing work.

The Beaufort wind scale for use on land (see below) provides indicators of prevailing wind conditions. Fixed or hand-held anemometers (wind velocity meters) should be used where additional accuracy is required.



Typical hand-held wind meter (anemometer)
(Image supplied by Kestrel Weather & Environmental Meters by NK)

Windforce number	Description of wind	Wind locally	Speed mph	Speed m/sec
0	Calm	Calm, smoke rises vertically.	1	0–1
1	Light air	Direction of wind shown by smoke drift, but not by wind or weather vanes.	1–3	1–2
2	Light breeze	Wind felt on face. Leaves rustle. Wind or weather vanes move.	4–7	2–3
3	Gentle breeze	Leaves and small twigs in constant motion. Wind extends light flags.	8–12	3–5
4	Moderate breeze	Wind raises dust and loose paper. Small branches move.	13–18	5–8
5	Fresh breeze	Small trees in leaf begin to sway. Little crested wavelets form on inland waters.	19–24	8–11
6	Strong breeze	Large branches in motion. Umbrellas used with some difficulty.	25–31	11–14
7	Near gale	Whole trees in motion. Becoming difficult to walk against the wind.	32–38	14–17
8	Gale	Twigs break off trees. Progress is generally impeded.	39–46	17–21
9	Strong gale	Chimney pots, slates and tiles may be blown off.	47–54	21–24



For detailed guidance on wind speed advice for different roof work refer to the HSE guidance *Health and safety in roof work (HSG33)*.

For further information refer to the National Federation of Roofing Contractors' guidance booklet *Roofing and cladding in windy conditions*.

2.4.3.3 Emergencies (including rescue)

Emergencies (such as a fire or gas leak) could occur either at ground level or at height, which requires that an emergency evacuation of the whole site, or evacuation of the roof where work is taking place, is carried out. For this reason the planning stage must ensure that safe access and egress will be available at all times. Depending upon the nature of the job, it may be necessary to have more than one access and/or egress route.

If the nature of the work and site layout necessitates that an asphalt or bitumen boiler, or other forms of hot work, have to be sited on the roof, then this must be considered as part of the fire risk assessment, local hot-work procedure and emergency arrangements.



For further information refer to the HSE guidance *Fire safety in construction (HSG168)*.

It may be necessary to deal with medical emergencies where someone becomes incapacitated at height through illness or injury and is unable to make their way back to ground level. In such circumstances, the local fire and rescue service may have to be involved to effect a safe rescue.

Consider emergency arrangements when selecting the method of roof access (for example, it may be easier to evacuate a casualty via a passenger hoist or stairway rather than ladder access).

Carrying out roof work will often involve the use of fall-arrest equipment (such as safety nets or safety harness and lanyard). Anyone who falls will have to be rescued promptly, particularly if they are suspended in a harness.

Where MEWPs are used to carry out the work it is essential that an effective emergency plan is produced, rehearsed and implemented. There must always be someone at ground level who is familiar with the ground controls and able to take action in the event of an emergency while a MEWP is in use.

The method of rescue needs to be proportionate to the risk and you should not rely on the emergency services.



For further information refer to Chapter D05 Fall arrest and suspension equipment.

2.4.3.4 Safe means of access and egress

The means of gaining access to height and safely working there will depend upon many factors (such as the nature of the roof structure, whether there is room to erect a scaffold or bring in a MEWP, and even the length of time that the job is expected to take).

Some common means of gaining access to height or actually working at height are listed below.

- Ladders.
- Mobile access equipment.
- Fixed or mobile towers.
- Stair towers.
- Independent scaffolds.

Before using an existing access and working platform (such as a permanently installed ladder or stairway) it should be inspected by a competent person to ensure its stability.

Additional temporary measures may be necessary to meet the requirements of the Work at Height Regulations for working platforms (for example, a parapet wall less than 950 mm high would require additional guard-rails).

2.4.3.5 Materials handling and storage

The planning process will involve taking decisions on the following.

- What roofing materials are required and where they can be stored safely at ground level.
- A safe means of transferring roofing materials to height and in what quantities, pack size and load (weight).
- Avoiding the overloading of any part of the roof including the integrity of steelwork members. Overloading can occur when materials are stacked on the roof prior to installation.
- The safe storage of sheet materials if they are to be stored for any length of time, particularly during windy weather.
- The safe distribution of materials around the roof during installation.
- The safe transfer of waste materials back to ground level.

Further notes on the safe stacking of materials on roofs are included later in this chapter (refer to 2.9 Handling and storage of roofing materials).

2.4.4 Stage 4. Carrying out the work

To a large extent, the risks of doing the job, the risk-control measures that will be necessary and therefore the way in which the work is carried out will depend upon the type of roof and nature of work to be undertaken.

Short duration means a matter of minutes rather than hours (up to a maximum of 30 minutes). It includes jobs such as brief inspections or minor adjustments (for example, minor adjustments to slipped slates).

Work on a flat roof is still dangerous, even if it lasts a short time. Appropriate safety measures remain essential.

It may not be reasonable to provide full edge protection for short duration work but something will need to be provided in its place. The minimum requirements for short duration work on a roof are safe means of access to and egress from the roof level and safe means of working on the roof (such as a properly constructed and supported roof ladder for sloping roofs, or a fall-restraint system for flat roofs) that prevents the wearer reaching a position where they could fall.

Mobile access equipment (such as scaffold towers or MEWPs) or proprietary access systems can provide a suitable working platform for some short duration tasks.

2.4.5 Stage 5. Post-completion information

The principal designer must ensure a health and safety file is prepared, developed and reviewed as the project progresses.

At the end of the project the principal designer must pass the file to the client. If the principal designer's appointment finishes before the end of the project, they must pass the file to the principal contractor, who must take responsibility for making any additional amendments and the hand over to the client at the end of the project.

The purpose of the health and safety file is to provide useful health and safety related information for anyone who, at a later date, has to carry out subsequent work (such as maintenance, cleaning, refurbishment or demolition).

In accordance with CDM it will be necessary for the roofing contractor to provide the principal designer (or principal contractor) with relevant health and safety information about the roof structure and materials used, as requested, for inclusion in the health and safety file.

2.5 Types of roof work

2.5.1 Flat roofs

On flat roofs, falls most frequently occur:

- from the edge of a completed roof
- from the leading edge where work is being carried out
- through openings or gaps
- through fragile material.



A flat roof is defined as any roof having a pitch of less than 10 degrees.



Flat roof with parapet wall barrier and skylight hazards

Safe access to the roof, and to any working place on that roof, must be provided and maintained. If there is no suitable parapet wall of 950 mm in height or similar barrier to stop anyone from falling, edge protection must be provided.

This may take the form of standard guard-rails and toe-boards or, providing nobody will approach the edge, a suitable physical barrier set back from the edge. All barriers should be positioned at least 2 m from a leading edge, to prevent workers from accessing danger areas where they would be exposed to a risk of falling. Barriers should also be secured to prevent them becoming displaced (such as in high winds). Warning signs should be displayed on the barriers.

Where works are to be undertaken that could result in materials or equipment falling onto people passing below, protective measures must be taken (for example, adding netting, close boarding or debris fans to scaffolds or establishing exclusion zones with fenced-off areas and warning signs to stop people from accessing the drop zone at critical times).

There could be circumstances when operatives need to work at exposed leading edges but it is not reasonably practicable to install guard-rails or other collective fall prevention measures. In these circumstances, it will be necessary to use personal work restraint systems or collective fall-arrest systems.

The most suitable type of fall-arrest system will be indicated by a risk assessment. Where safety nets, airbags or other soft-landing systems are used, they will provide collective fall protection for anyone working above them who falls.

The last resort would be the use of a personal fall-arrest system, such as a safety harness with a lanyard clipped to a suitable anchor point or a horizontal running line. The effectiveness of this system depends upon the training and personal discipline of operatives. Adequate supervision is required to ensure the system is effective (such as the operatives actually clipping on).

Employers should consider work restraint systems where the lanyard should be sufficiently short or can be adjusted so that it prevents the wearer from reaching a position from which they could fall.

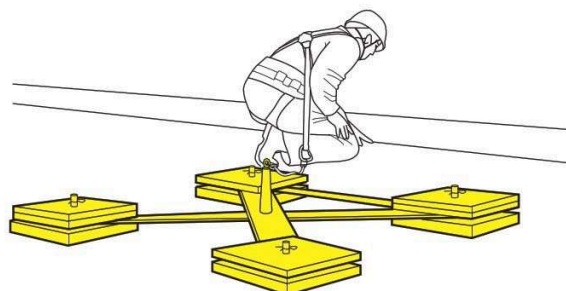
When a safety harness and lanyard with an inertia reel device is used, consideration must be given to the position of the anchor point, which should be above the head-height of the user.

It is important to note that, when using this form of fall-arrest system, the anchor point and self-retracting lifeline (SRL) are as high as possible to ensure that the fall factor, which provides an indication of the length and severity of fall, is kept as low as possible.

As a guide, when using a 2 m length lanyard, the fall distance required (to provide adequate clearance to the fallen operative) is 5.2 m when attached above head height, 5.75 m when connected at shoulder height and 6.75 m when attached at foot level. Where the anchor point is at ankle level, for example, there will be more slack in the lanyard so the fall will be further before it is arrested.

Inertia reel blocks are designed to be used vertically with a restricted horizontal movement. If the inertia reel moves through angles it will become less effective. Users should refer to the manufacturer's instructions to ensure inertia reels are used correctly. If not, they can create a pendulum effect (swing fall risk) and, as well as not stopping the fall, they can expose the user to additional hazards.

The use of this type of equipment has become more common as it is not as restrictive as a fixed line system.



Temporary anchorage for a fall-restraint system (Photo © NASC)



For further information refer to Chapter D05 Fall arrest and suspension equipment.

Sometimes, guard-rails have to be moved or removed to enable work to be undertaken. Consider using proprietary edge protection systems that are designed to allow adjustments without compromising the guard-rail protection (for example, raising a guard-rail post to allow access to the roof surface). If temporary removal of a guard-rail is unavoidable then the following precautions should be followed.

- An equally effective safe system of work must be in place and maintained, which will prevent falls of persons or materials.
- The guard-rails must be replaced or re-erected as soon as practical.

Some flat roofing systems will involve the use of various chemicals or hot works and fumes or solvents may be present. Consideration should be given to the Control of Substances Hazardous to Health Regulations 2002 and the manufacturer's information, as well as the significant possibility of fire, explosion or burns from hot bitumen.

2.5.2 Traditional pitched roofs

On traditional pitched (sloping) roofs, most falls occur:

- from the eaves or ridge, by slipping down then falling from the roof
- during truss erection (falling into the structure)
- from unprotected gable ends
- through fragile roofing materials, particularly fragile roof lights.



A pitched roof is defined as any roof having a pitch of more than 10 degrees.

Work on pitched roofs should only be carried out in the following circumstances.

- By persons who are physically capable, trained and competent.
- Using roof ladders or a temporary work platform equipped with guard-rails and toe-boards, as necessary, and securely fixed to prevent it slipping. (The use of homemade or improvised roofing ladders should be prohibited.)
- Providing that either a suitable catch barrier or a working platform with guard-rails is erected at the eaves and verge of the building.

These requirements apply to any work on a pitched roof, including access to, and egress from, other workplaces.

A steeper roof pitch increases the risk of slipping down the roof surface. The European Standard BS EN 13374 provides guidance and a specification for temporary edge protection depending upon the slope of the roof and potential fall distance.

Working platforms should be provided where the pitch of the roof prevents a secure foothold. Alternatively, work restraint systems or rope access techniques may need to be considered.



Traditional pitched roofs

WORKING AT HEIGHT

2.5.3 Curved roofs

Curved roof surfaces (such as a barrel-vault structure) may be used as a working platform, but this will depend upon the radius of the curve, as to whether a secure foothold can be safely maintained (similar to steeper pitched roofs). The more acute the curve the greater the risk of slipping and falling down the roof.

Therefore, it is often necessary to employ rope access techniques (such as work positioning or abseiling). It is essential that any such system incorporates lines secured to the apex or other suitable anchor point and a self-locking device to prevent the uncontrolled descent of anyone who loses their footing.

Additionally, proprietary rubber steps are available that follow the exact contour of a curved roof, with sections being joined to extend the overall length. These steps must be regarded as a foothold only, as they are not equipped with a handrail. They must therefore only be used in conjunction with another form of fall protection (such as a fall-arrest block).

Where there is a risk of falling through a curved roof, fall-arrest safety netting or similar protection should be installed. Special measures are required when rigging safety nets to curved type roofs.

2.5.4 Profiled roofs

These range from a single asbestos cement sheet on an outside toilet to the latest continuously extruded system formed in situ to cover large portal frame buildings. The systems will differ in complexity and range from single skin and multiple skin (build-up type) through to composite factory-produced sheeting units.

New build profiled installation will normally be carried out over a system of safety nets that provide fall protection.



For further information refer to Chapter D05 Fall arrest and suspension equipment.

Other systems include temporary leading edge devices (such as stages with guard-rails) and trolley systems that are designed to run along purlins. However, installation, use and moving of these systems can be high risk and will normally require the use of additional fall protection measures (such as safety nets or a safety harness system). The HSE recommends that trolley systems should only be used where there is no safer option.

Designers who specify roofing materials should consider fragility and provide precise information on fixing configurations and load-bearing capacity, so that contractors can prepare safe systems of work.

Composite roof sheet systems are often immediately load bearing, independent of fixings. However, safe systems of work should be developed for handling these heavier sheets.

If a built-up roof system is being installed, the inner sheet or liner tray must be securely fixed before it can be stood on. Metal profile roof sheets are therefore still fragile until they have been fixed.

Furthermore, many roof-light assemblies, which are often installed as part of a profiled roof system, may also be fragile. This will necessitate proprietary work platforms (such as stagings) that are used to enable safe access and an additional means of fall protection (such as safety nets).

Edge protection should be provided in accordance with BS EN 13374 (practical method of providing temporary edge protection for working on roofs). Any working platform must comply with the Work at Height Regulations and be, ideally, a minimum of 600 mm wide with handrails on one or both sides, depending on whether a fall can occur.



Edge protection on a steel structure

2.5.5 Trussed roofs

The design of safe access into a roof truss system should be undertaken by the truss designer or architect, and this information passed to the contractor. This will then form part of the overall risk assessment.

A safe working platform around the roof perimeter should be erected and safe access provided where required within the trusses.

The most common fall protection systems typically used for internal falls when constructing a trussed roof include the following.

- Proprietary decking systems.
- Interior birdcage scaffolding.
- Boarding out the bottom chord of the trusses (as long as they are stable and capable of taking the imposed loads).
- Soft-landing systems (such as airbags or bean/foam bag types).

Safety nets can be used, providing that a safe clearance distance can be achieved below, and that there is suitable anchorage.

2.5.6 Solar panel installation and maintenance

The installation and maintenance of solar energy photovoltaic (PV) panels on roofs presents the same inherent risks to health and safety as roof work itself and, therefore, the same levels of protection need to be adopted. Consideration should be given to the following points.

- Safe access to the roof surface (such as scaffolding work platforms or edge protection) for fixing rails, brackets, PV panels, and so on, as a priority before resorting to personal fall protection (such as harnesses and work restraints).
- Working on or near fragile roofing materials.
- Raising and handling heavy PV panels at height.
- Future access for repairs or maintenance.

[www](#) For further information refer to *Solar panel installation (GS001)*.

Magistrates issue £250,000 fine following worker's fall from roof

A solar panel installation company was fined after failing to control risk and supervise work, resulting in a 49-year-old worker falling more than 3.5 m through a void in a roof at a school. The man suffered serious injuries, including bleeding on the brain, a ruptured spleen and fractured ribs.

A Magistrates' Court heard how he was one of several workers contracted by the solar panel installation company to undertake roof works. HSE investigations found that the company failed to ensure physical guarding was in place to prevent the worker and his colleagues from falling through voids in the roof. The inquiry also found that the company failed to appropriately supervise the work.

The solar panel installation company pleaded guilty to breaching Regulation 4(1) of the Work at Height Regulations 2005 and was fined £250,000 and ordered to pay costs of £12,073.14.

Speaking after the hearing, an HSE inspector said: 'The company failed to properly organise the work and make sure the workers knew what safety measures were needed, as well as providing appropriate supervision to ensure work was not undertaken without the measures being put in place. As a result, serious harm was caused to one worker and others were put at serious risk'.

(Source: HSE.)

2.6 Fragile roofs and surfaces

All roofs, once fixed, should be treated as fragile material until a competent person has confirmed otherwise.

Falls through fragile surfaces account for over 20% of all fatal injuries of falls from height.

The following are likely to be fragile.

- Cement sheeting (non-reinforced sheets, irrespective of profile type, for example asbestos).
- Roof lights (difficult to see in certain light conditions or hidden when painted).
- Liner panels on built up sheet roofing.
- Metal sheets, where corroded.
- Glass, including wired glass.
- Chipboard or similar materials, where rotted or not sufficiently supported.
- Other materials, including wood wool slabs, slates and tiles.

As far as the strength of materials is concerned, the appearance of fragile roofs is often misleading. Surface coatings, dirt or moss may give a false appearance of soundness to glass, plastic, asbestos, and so on. Even if the roof is clad in a load-bearing material, roof lights can often be fragile. Asbestos and various plastic materials are particularly brittle and may shatter without warning.



The CITB video: *Fragile roofs - Fragile lives* shows the devastating effects that falls can have

Falls from roofs

- A 50-year-old roofer was killed when he fell from the edge of a pitched roof. He was understood to have been carrying out extensive repair work following storm damage. There was no edge protection and the access ladder was not secured.
- A 60-year-old employee was killed when he fell through a fragile roof while installing a ventilation duct for a spray booth.
- A self-employed builder, aged 52, fell through a fragile roof light while trying to repair damage to the asbestos cement roof of an industrial unit.

WORKING AT HEIGHT

Effective precautions are required for all work on or near fragile surfaces, no matter how short the duration, whether the work concerns construction, maintenance, repair, cleaning or demolition.

Ideally, another way of carrying out the job that does not require anyone to work on or near to fragile roofing materials should be found.

Fragile materials should be protected when anyone passes by or works near to them. They should be securely covered and a warning notice displayed if people are working within 2 m of the area. Appropriate precautions should be taken when installing protection (such as the use of netting, birdcage scaffold or a harness system).

Sometimes it will not be reasonably practicable to provide such protection, usually if the proximity to fragile materials is irregular and of short duration (such as a matter of minutes).

Safety harnesses will usually be the most appropriate solution and may be used in conjunction with any permanently installed running line systems.

Boundaries can be established using barriers to identify safe areas within the workplace and routes to and from them. The safe area boundary should be at least 2 m from the nearest fragile material and leading edge.

It does not need to comply with full edge protection standards, but there should be a continuous physical barrier (a painted line or bunting is **not** acceptable), and discipline is essential to ensure that everyone stays inside the safe area at all times.

Where walkways with a handrail are not an integral part of the roof structure, a safe system of work must be devised.

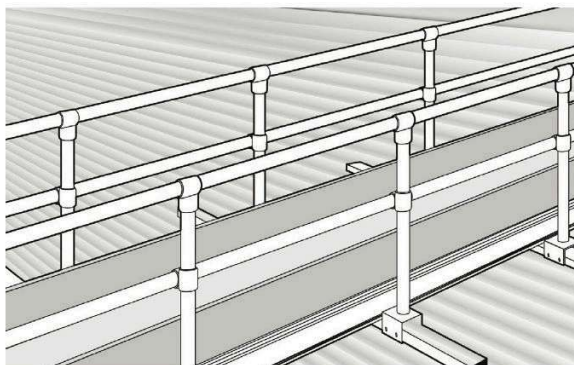
If it is necessary to pass across a fragile roof, a roof-board complete with a guard-rail should be used to spread the weight and provide a good handhold. Depending upon the job, it may be necessary to use more than one roof-board: one to support the person, whilst the other is moved to a new position.

In such cases additional fall protection may be necessary (such as fall-arrest netting or a safety harness system) if there remains a risk of a fall through a fragile roof surface.

Walkways near fragile surfaces (in valleys, parapets, gutters or channels) must be provided with suitable guard-rails or, if not, the fragile surface should be overlaid with a load-bearing material to prevent the possibility of anyone falling through.

Warning signs should, so far as is reasonably practicable, be fixed at all approaches to roofs constructed with fragile materials. This is a requirement of the Work at Height Regulations. Such signs must comply with the Health and Safety (Safety Signs and Signals) Regulations 1996.

Where such signs cannot be or are not fixed in place, it is essential that the presence of fragile material is identified in advance and those doing the job are made aware of it by other means, such as during risk assessment/method statement briefings, task safety briefings, or toolbox talks.



Temporary working platform



Rooftop walkway with guard-rails preventing access to a leading edge of fragile materials (Image supplied by Kee Safety Ltd)



Trying to walk the line of the bolts or purlins on a roof is dangerous and must not be attempted.



Refer to the HSE publication *Health and safety in roof work* (HSG33) (paragraphs 170–202) and *Fragile roofs* (GEIS5) for details of the dangers presented by fragile surfaces.



Fatal fall through a skylight

A scaffolding firm was ordered to pay more than £100,000 in fines and costs following the death of an employee who plunged 13 m through the roof of a warehouse.

The employee stepped onto a skylight, which broke and gave way. There were around 80 fragile skylights, each measuring 1 m x 2 m, on half of the roof.

However, the company failed to arrange for covers to be put on the skylight nearest to where the employees were working.

! Things to consider

Avoidance. Plan and organise work to keep people away from fragile surfaces, so far as possible (for example, undertake work from below the surface in a MEWP).

Control. Working on or near fragile surfaces requires a combination of staging, guard-rails, fall restraints, fall arrests and safety nets slung beneath.

Communication. In addition to the controls, fix warning notices to the approach of any fragile surface. Workers must be trained, competent and instructed in the use of the precautions required.

Co-operation. On business premises contractors should work closely with the client and agree the arrangements for managing work.

2.7 Managing asbestos in roof work

Asbestos-containing materials have been widely used in UK roof construction for many years. It is a legal requirement of the Control of Asbestos Regulations 2012 (CAR) that anyone liable to disturb asbestos during their work, or who supervises such employees, receives the correct level of information, instruction and training. The level of training required will be determined by the type of work being carried out.

The HSE maintains a list of accreditation bodies for training organisations that can deliver training of this nature.



The HSE has produced *Asbestos essentials*, a task manual for building, maintenance and allied trades on how to safely carry out non-licensed work involving asbestos.

The method statement must be written by someone who is deemed competent with regard to asbestos and who understands the requirements of CAR. All asbestos-containing materials removed from roofs will be classed as hazardous waste (special waste in Scotland) and will need to be disposed of by a registered waste carrier at a licensed facility under a hazardous waste consignment note.

Under CAR an employer must assume all materials are asbestos-containing unless they can prove otherwise (for example, by reference to the asbestos survey undertaken to meet the requirements of *Asbestos: the survey guide* (HSG264) and by extension of CAR). Whilst it is not difficult to recognise asbestos cement roof sheets, determining if roofing felt, soffits or insulation boards contain asbestos is normally more problematic.

All commercial clients have a duty to provide information about the presence of asbestos and should be able to give designers and contractors a copy of their asbestos register and/or asbestos survey report. Domestic clients have no such obligation, and it will be the contractor's responsibility, prior to work commencing, to identify any asbestos-containing materials present. This may require a survey in line with HSG264 (identified above). All asbestos removal work must be undertaken, in accordance with CAR, by those competent to do so.



For further information refer to Chapter B09 Asbestos.

2.8 Access arrangements

2.8.1 Stair towers

Stair towers are an effective and safe means of access. They help speed up access, production and enable the workforce to carry materials and equipment far more safely than ladder access.

Many companies now default to stair towers, unless a risk assessment deems that ladder access is suitable for the situation (such as short or low-risk access).

2.8.2 Ladders

Ideally, access to a scaffold or roof will be via a purpose-built stair tower. However, where the decision is taken to use a ladder, the scaffold should be constructed to incorporate internal ladders and protected ladder traps.

If an external ladder is used it should be positioned along the face of the scaffold rather than at right-angles to it. Ladder gates should be installed at access points.



A typical site stairway tower
(Photo © NASC)



External scaffold access with safety gate
(Photo © NASC)

WORKING AT HEIGHT

A ladder must not be used as access to, or egress from, a workplace unless it is reasonable to do so, taking into account the work being carried out, its duration and the risks to the safety of any person arising from the use of the ladder.



Competence is vital

Demonstration of competence is a vital requirement of the Work at Height Regulations, and successful completion of the Ladder Association's industry standard training course contributes significantly to providing proof of competence when working with ladders and stepladders.



For further information on this course, and the regulations, refer to the Ladder Association website and Schedule 6 of the Work at Height Regulations.

2.8.3 Landing areas (rest platforms)

Landing areas must be provided for ladders where the vertical height is over 9 m, must have adequate dimensions and must be provided at suitable intervals. All landing areas must be provided with the following.

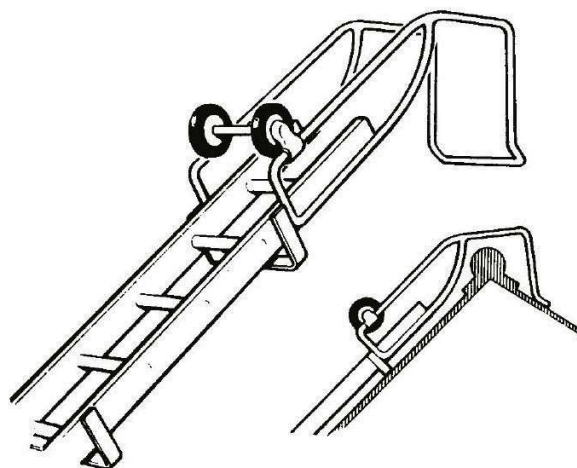
- A guard-rail at a height of not less than 950 mm.
- An intermediate guard-rail.
- A toe-board.
- A gap not exceeding 470 mm between the toe-board and guard-rail, or between any two guard-rails.

2.8.4 Roof ladders

Personnel involved in roof work should not walk directly on slates or tiles on sloping roofs, as these do not normally provide a suitable foothold, especially when wet.

Roof ladders, designed for the purpose, should be secured by means of a ridge hook over the ridge tiles, bearing on the opposite roof pitch.

The ridge tile or gutter should not be used to support roof ladders, as they can become accidentally displaced.



A proprietary roof ladder

2.8.5 Edge protection

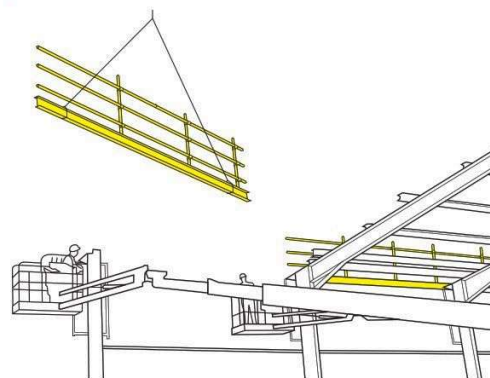
Falls from the edge of the roof should be prevented by perimeter scaffolding or an edge protection system fixed to the building structure. Designers of new buildings should consider how edge protection may be secured to the building for construction and future maintenance.

Edge protection to prevent the fall of people and materials should be provided in accordance with the Work at Height Regulations and BS EN 13374 for construction and maintenance work. This British and European standard is not generally suitable for the public, other workers, vehicles or mobile plant equipment or bulk materials (such as packs of roofing materials). The classification of edge protection systems is primarily based on the gradient of the surface and the potential fall distance down the roof for which they are to provide protection.

The performance requirements for the various classes are detailed within the standard BS EN 13374. These requirements are briefly summarised below and in the images on the following page.

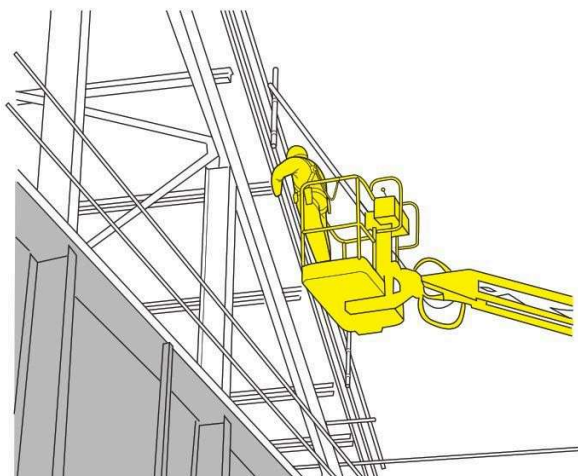
Class A provides protection to flat surfaces and slopes of generally up to 10 degrees. It provides resistance to static loads and is based on the requirements to support a person leaning against, walking beside, and possibly stumbling against the edge protection.

Class B provides protection to flat surfaces and slopes of generally up to 30 degrees, and to even steeper slopes with short slope lengths. It provides resistance to both static and low dynamic loads and is based on the requirements to support a person leaning against, walking beside, possibly stumbling against and sliding down a sloping surface towards the edge protection.

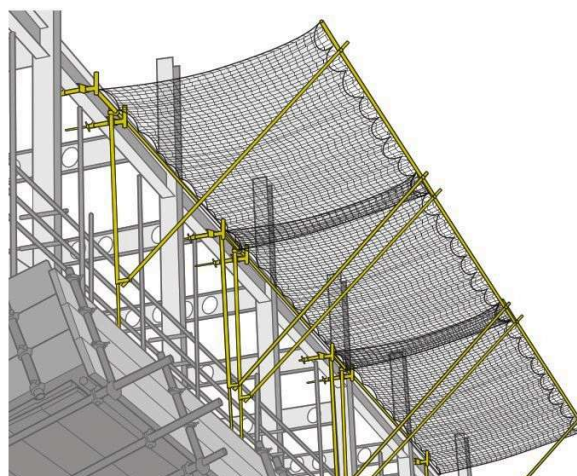


Temporary edge protection pre-fixed to steelwork before installation
(Photo © NASC)

Class C provides protection to steeply sloping surfaces of generally up to 45 degrees, and up to 60 degrees for 5 m slopes. It provides resistance to high dynamic loads only and is based on the requirements to contain a person sliding down a steeply sloping surface. All edge protection systems should be adequately secured so that they cannot become accidentally displaced in their intended use (loading as per BS EN 13374 or, in the case of scaffolding, BS EN 12811).



Scaffolding edge protection installed from a MEWP (Photo © NASC)



A proprietary safety net fan system provides fall protection, enabling perimeter guard-rails to be erected above without MEWP access (Photo © NASC)

Safe systems of work must be established for the installation and removal of temporary edge protection. Where possible, edge protection installed to the building structure should be accessed by a MEWP. Where MEWP access is not practical, collective fall-arrest fans or, as a last resort, fall-restraint systems should be used.

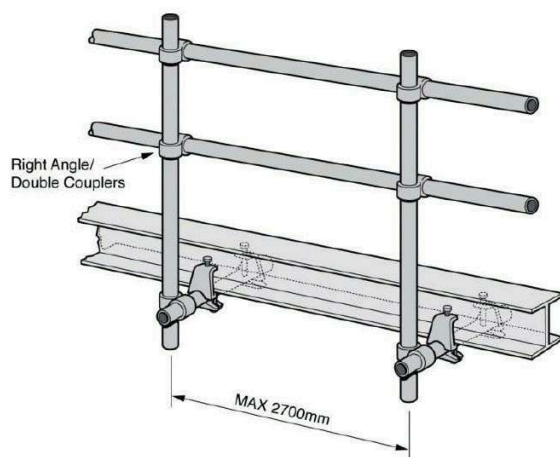
Where scaffolding is used with a working platform, guard-rails and toe-boards to provide roof edge protection, it should be positioned close to the eaves to minimise the potential fall distance to the platform. This should be ideally no more than 300-450 mm below the eaves or roof edge.

However, sometimes access may also be required to the eaves (for example, fascia and soffit) where the fall distance may be slightly greater (up to 600 mm) but avoids unnecessary scaffold adaptations.

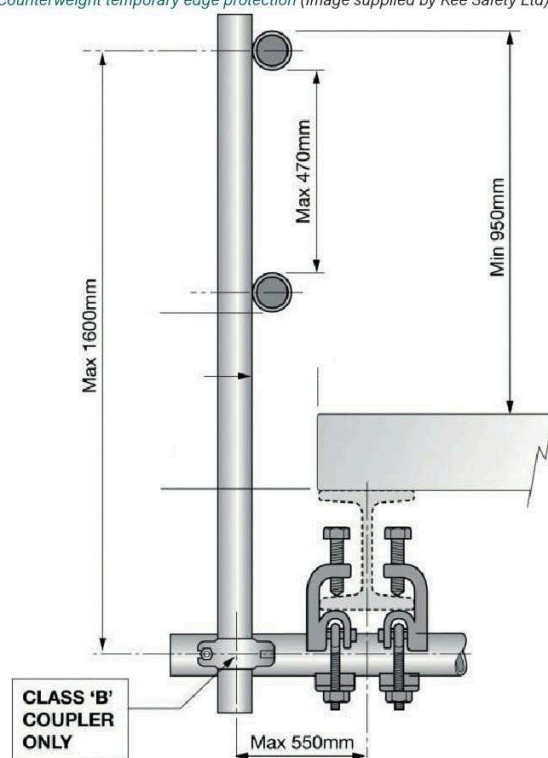
In such cases the increased risk of injury must be assessed so that the fall distance is minimised and consideration given for safe access to and egress from the roof surface and the possible need for an additional guard-rail for increased fall protection (for example, internal guard-rails to the eaves or an additional, third guard-rail above the main platform guard-rail).



Counterweight temporary edge protection (Image supplied by Kee Safety Ltd)



Temporary edge protection on open steelwork (Class A)



WORKING AT HEIGHT

02

It is recommended that the platform should be wide enough to provide a suitable catchment area should someone fall from a sloped roof. Ideally the platform should extend 2 m past the work area on the roof, or guard-rails should be installed on the roof to contain the workers within the protection offered by the scaffolding. Special consideration should be given in the following situations.

- When workers can encroach within 2 m of the verge, guard-rails or gable-end scaffolding may be required.
- If there is a risk of workers falling down the opposite side of a ridge and unprotected eaves. Edge protection to the opposite eaves should be provided or a work restraint system used.
- When workers can encroach within 2 m of an opening or fragile surface within the roof surface.

2.8.6 Access to chimney stacks

Construction and maintenance of chimney stacks should always be undertaken from a properly constructed working platform with safe access. Traditionally, scaffolding to chimney stacks offers the safest option. However, risks involved during erection and dismantling also need to be considered.

There are various purpose-designed, lightweight stagings and access systems available for work on or around chimney stacks. MEWP access may also be a practical solution for short duration or inspection work before resorting to ladders. Consideration should be given to the roof structure and load imposed by the access equipment. If in doubt engineering advice should be sought.



For further information on scaffolding refer to Chapter D04.

2.8.7 Powered access

Powered access covers a wide range of mechanical access equipment, such as those listed below.

- MEWPs.
- Mast climbing work platforms (MCWPs).
- Suspended cradles.
- Personnel carrying platforms.

Mobile access platforms are often used as an alternative to ladders, scaffolds and cradles.

Each of these types of equipment can be suitable for carrying out particular types of roof work (for example, using a telescopic boom MEWP (also known as a cherry picker) might be a suitable and safe way of accessing a job that would otherwise require someone having to cross a fragile roof). This is particularly important for inspection or short duration maintenance work. Truck-mounted MEWPs now have sufficient reach to enable inspection of many building types to be made without having to directly access the roof. The use of drones is increasingly being used to carry out a range of activities including inspection and surveys for working at height.

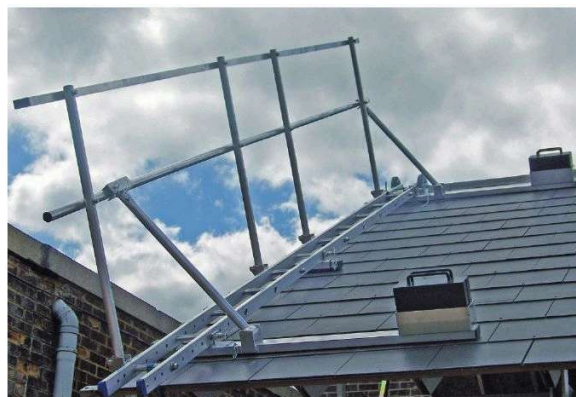


For further information on drones see Chapter C05.

2.8.7.1 Mobile elevating work platforms (MEWPs)

- All MEWPs must conform to *Mobile elevating work platforms. Design calculations. Stability criteria. Construction. Safety. Examinations and tests* (BS EN 280).
- MEWPs must be operated by authorised and trained persons who hold the appropriate International Powered Access Federation (IPAF) powered access licence (PAL) for the type of MEWP to be used.
- Operators must be familiar with the model of MEWP and maintain a log book to record experience.
- Safety harnesses with fall restraint lanyards must be used in boom-type MEWPs (except when working over or near water), to prevent operatives from accessing a position where they could fall.
- Only use suitable anchor points, as specified by the manufacturer. Never use the guardrails as an anchor point.
- Fall restraint lanyards must be adjusted to the shortest practical length necessary for the platform, ideally no more than 1.5 m long.

Note: adjustable or suitable length fall-arrest lanyards can be used in a fall restraint mode, as energy absorbers will not deploy until a load of 3 kN is applied.



Edge protection on a sloping roof

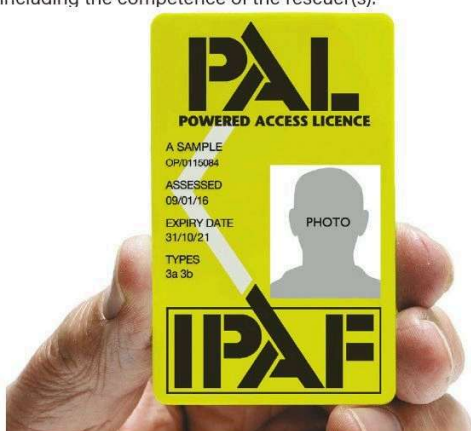


Example of a chimney stack and access scaffold (Photo © NASC)

- When working in a MEWP harnesses should not be used in a fall-arrest scenario as the dynamic loads imposed could be beyond the capacity of the MEWP and cause overturning or collapse.
- Never exceed the safe load rating of the MEWP. Specially designed devices are available for handling materials within a MEWP (such as cladding installation systems). Seek advice from the supplier.
- Never use a MEWP as a hoist, crane or prop, unless it is designed for the purpose.
- MEWPs must be used in accordance with the *Operators' safety guide* (IPAF) and *Safe use of MEWPs – Code of Practice* (BS 8460).
- Care should be taken to avoid trapping and crushing people in the platform.
- When operating a MEWP in a high-risk area of entrapment (such as near overhead steel beams), equipment that has a secondary guarding device installed must be selected.
- MEWPs must be inspected and maintained in accordance with the manufacturer's recommendations and subject to thorough examination as per the Lifting Operations and Lifting Equipment Regulations 1999 (LOLER) every six months.
- Planning for emergencies must include how rescue and recovery from a MEWP is to be carried out, including the competence of the rescuer(s).



Scissor lift type MEWP in use



IPAF powered access licence



IPAF operators' safety guide



For further information refer to the Strategic Forum Plant Safety Group Best Practice Guidance for MEWPs – Avoiding trapping/crushing injuries to people in the platform.

2.8.7.2 Mast climbing work platforms (MCWPs)

MCWPs are not commonly used or suitable for roof access or edge protection; their primary function is to provide a movable façade access to buildings that, when repositioned, will leave the leading edge exposed. With this equipment, the platform or cradle rises up one or more static masts in a similar way to a hoist, providing a temporary work platform at height, usually on the side of a building. Some procedures to be observed in the erection of mast climbers are described below.

- Installers must be trained, in accordance with the manufacturer's instructions.
- Make sure there is a system in place to identify a loss of mechanical integrity in each drive unit. Check each unit is fitted with a mechanical device, such as a centrifugal brake.
- MCWPs should be operated by authorised and trained persons who hold the IPAF PAL special category for MCWPs.
- MCWPs must be inspected daily, before use, by the user.
- MCWPs must be regularly inspected by a competent person and records maintained of the inspection. They must be thoroughly examined every six months.
- The safe working load and the permitted number of persons allowed on the platform at each configuration must be clearly marked on the MCWP. They should not be used as a substitute for using stairs or a passenger lift for travelling to higher levels.
- Planning should consider the interface with building features that could impede the working envelope of the MCWP and present trapping or shear points to the users or residents of occupied buildings (such as opening windows and balconies).
- All MCWPs must conform to the *Code of Practice for the installation, maintenance, thorough examination and safe use of mast climbing work platforms* (BS 7981) and *Lifting platforms. Mast climbing work platforms* (BS EN 1495).



For further information on MEWPs and MCWPs, visit the IPAF website.

WORKING AT HEIGHT

2.8.7.3 Suspended cradles

This section covers suspended working platforms, commonly referred to as cradles. As with MCWPs, cradles are not commonly used or suitable for roof access or edge protection, as their primary function is for façade access.

- All proprietary cradle systems must conform to *Safety requirements for suspended access equipment. Design calculations, stability criteria, construction. Examinations and tests* (BS EN 1808).
- Temporary suspended cradles and scaffolds must conform to the *Code of Practice for the planning, design, setting up and use of temporary suspended access equipment* (BS 5974).
- The design, inspection, testing and use of suspended cradles must be in accordance with the Specialist Access Engineering and Maintenance Association (SAEMA) guidance.
- Installers of proprietary cradle systems must be trained in accordance with the manufacturer's instructions.
- Users of suspended cradles must use personal fall protection equipment and be attached to a designated anchor point within the cradle or line system.
- The use of permanently installed cradle systems for access during building construction, designed for future maintenance use, may be a practical option. However, consideration should be given to the nature of use and advice should be sought from the manufacturer.



Permanently installed cradle system for future building maintenance used during construction operations

www For further information, visit the SAEMA website.

2.8.7.4 Personnel-carrying platforms

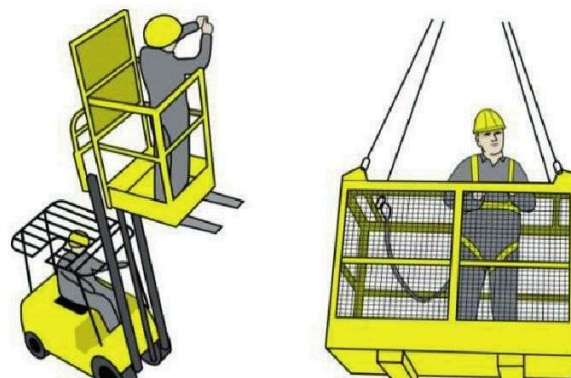
Personnel-carrying platforms (or baskets) are not powered access equipment. However, they are designed to be used with cranes or lift trucks and, when combined, provide a means of powered access. Consideration should be given to a MEWP, or similar purposely-designed access equipment, before resorting to personnel-carrying platforms.

The use of baskets with rough terrain or telescopic type lift trucks in construction increases the risks to users, and therefore safer alternatives (such as MEWPs) should always be considered first.

- Personnel carrying baskets provide specific hazards, as the user is reliant on the operator of the crane or lift truck. Therefore, there must be an effective means of communication and a signaller used, as per other lifting operations.
- The use of a personnel carrying platform must be planned, managed and operated, the same as any lifting operations under LOLER.
- The type of personnel carrying platform must conform to *Cranes. Equipment for lifting of persons. Suspended baskets* (BS EN 14502).
- The lifting device (crane or lift truck) must be suitable and have the correct load rating.
- The personnel carrying platform must be slung and/or secured to the lifting device, as per the manufacturer's instructions.
- Passengers in the personnel carrying platform must use personal fall protection equipment and be attached to a designated anchor point within the platform or the lifting equipment (for example, a crane hook).
- Due consideration must be given to the thorough examination requirements of LOLER when mobile work equipment is used to lift people, including the accessory itself: six months for lifting equipment and any associated accessories used to lift people, and six months for all lifting accessories and 12 months for all other lifting equipment.

For further information refer to:

- Chapter D05 Fall arrest and suspension equipment
- Chapter C05 Work equipment and hand-held tools
- Chapter C07 Lifting operations.



Lift truck mounted and crane slung personnel carrying platforms

2.8.8 Safety nets

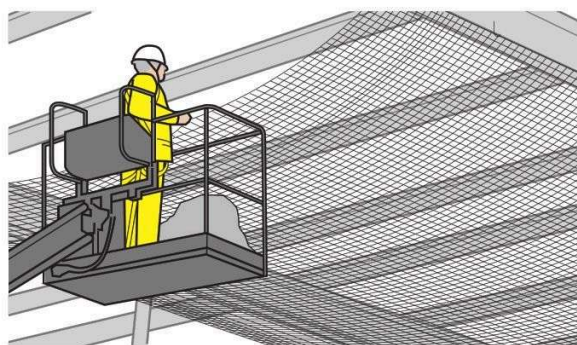
For most industrial and commercial roofing and flooring construction, properly rigged fall-arrest safety nets and perimeter edge protection is the preferred solution. Nets provide workers with maximum freedom of movement, as long as they do not encroach beyond the protected area.

- Nets must be supplied in accordance with BS EN 1263 Part 1 and rigged in accordance with BS EN 1263 Part 2 and BS 8411 (*Code of Practice for safety nets on construction sites and other works*).
- Prior to installation and use on site, **all** safety nets must have been patch tested within the previous 12 months (unless they are new, i.e. less than 12 months old).
- All nets must have a tag displaying:
 - the net's serial number
 - date tested
 - date of next test, and
 - confirmation that the net meets the minimum test criteria set by the manufacturer.
- Only competent and trained riggers are permitted to rig, alter or de-rig safety nets, ideally in accordance with the *Fall arrest safety equipment training* (FASET) scheme, or equivalent.
- Anchorage for safety nets must be capable of withstanding 6 kN of force. Therefore safety netting must be tied to a suitable structure. Never fix nets to scaffolding unless it has been designed for the purpose.
- Ensure the most suitable method of access is used for rigging and de-rigging safety nets, in accordance with the FASET safety guidance notes.
- Remote methods of rigging and de-rigging should be used wherever possible.
- Rescue plans must consider recovery of a casualty from the safety nets.
- Nets can be rigged with debris netting for light debris containment.
- Net repairs must be tagged by a competent person, in accordance with FASET recommendations.
- Never store materials below safety nets that reduce clearance distances.
- Nets must be inspected by a trained safety net rigger or inspector, before use, every seven days, and after any event likely to jeopardise the safe condition of the nets.
- Nets should have a further inspection by a competent person every seven days after handover, if the nets continue to be worked above.
- A safety net inspection register must be maintained.

For further information refer to the following:



- **FASET website.**
- ***The Blue book - Recommended practice for the use of safety nets for roof work*, published by the Advisory Committee for Roofsafety.**



Typical method of rigging a roof safety net via a MEWP
(Photo © NASC)

2.8.8.1 Working platform nets

Tensioned netting systems are being used more commonly within the construction industry for roofing operations and to enable access to difficult workplaces, for example under deck areas of structures such as bridges, piers, jetties and offshore platforms. Tensioned netting systems must be subject to a specific design scheme.

- Installation and use must be properly planned, risk assessed and a detailed method statement produced.
- They must be installed, inspected, used and maintained in accordance with FASET recommendations and specific design.
- The design must determine the maximum loading and the load imposed on the supporting structure.
- A methodology for the safe use must be established for the management of this work at height access system as part of the specific design of the scheme.
- As a minimum, working platform nets should be inspected to the same standard as fall-arrest safety nets.

2.9 Handling and storage of roofing materials

Care must be taken at all times when stacking bulk material on or at roof level. Attention should be given to the following points.

- Size of the load involved.
- Types of material involved.
- Methods of raising the load, whether manually or mechanically.
- Means of communication (signals) and the competence of the slinger or signaller.
- Position authorised for stacking materials.
- Distribution of the loads (loading plan).
- Maximum load or stack size.
- Loading limitations of the roof.
- Adequate support or packing to the truss.
- Provision of exclusion zones at ground level.
- Careful stacking of materials on roofs.
- Protection of the existing roof surface and any weatherproofing.
- Prevention of the displacement of loads that should be secured against:
 - the wind, especially split bundles and sheets
 - sliding down sloping roofs (sheet stop).

Manual handling of materials needs to be considered by roofing contractors as part of the risk assessment process, especially heavy or unwieldy sheeting materials and the increased risk associated with adverse weather conditions.



For further information refer to *Health and safety in roof work (HSG33)* and the NFRC guidance for handling roofing sheets and recommended safe wind speeds.

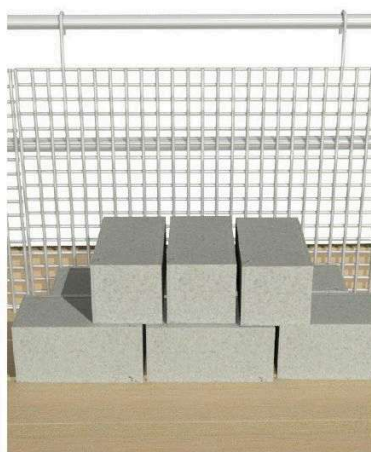
Mechanical handling equipment should be used where possible for handling bulk materials. Specially strengthened sections of scaffolding or loading bays may be required.

Where roofing materials are being stripped from an existing roof and stored on a scaffold for reuse, they must be stacked safely, not overloading the structure, and there should be sufficient space for working and safe passage.

The risks from falling materials and other objects in roof work must be considered and suitable controls established to prevent and protect against this.

Where the risk of falling objects cannot be completely eliminated, the risk must be managed. Some precautions are shown below.

- Toe-boards on working platforms.
- The use of brick-guards.
- Protective fans (*refer to Chapter D04 Scaffolding*).
- Containment sheeting and debris netting, including overlay debris netting for fall-arrest netting.
- Pedestrian gantries with crash decking.
- Debris chutes for removal of waste materials from the roof.
- Segregating the area below as a designated danger area and preventing unauthorised access with physical barriers and displaying appropriate warning signs.



Example of a brick-guard (Photo © NASC)



Example of a debris chute (Photo © NASC)

2.10 Other considerations

This section is primarily focused on roof work. However, work at height and associated risks are present in a wide range of construction, refurbishment and maintenance activities. Employers must ensure all work at height is properly planned and managed in accordance with the Work at Height Regulations.

2.10.1 Danger areas

Under the Work at Height Regulations, there is a specific legal requirement to ensure the safety of those working at height (from falls), and others who may be working or passing below (from falling objects), by the creation of danger areas, either around the high-level workplace and/or below it.

This is achieved by preventing unauthorised access into any danger area, so far as is reasonably practicable, by the use of barriers and appropriate signs or in some cases designated personnel to control access in and around the danger zones.

Where designated personnel are used to control danger areas it is essential that an effective means of communication is established between themselves and those working at height.

2.10.2 Personal protective equipment



In the context of roof work, personal protective equipment (PPE) includes items of fall-arrest equipment (such as safety harnesses and lanyards).

In deciding which type to issue, the employer must take into account the risk that the PPE is being used for, and also ensure that the PPE will fit the wearer and allow them to work comfortably.

The employer must ensure that employees have been given adequate and appropriate information, instruction and training to enable them to understand the risks, the purpose of the PPE and the manner in which it is to be used.

Whilst the employer must take reasonable steps to ensure that any PPE supplied is used, the employee, in turn, must use the PPE provided in accordance with instruction and training given and follow the procedures for reporting loss or defects to their employer.

2.10.3 Presence of bats

Bats commonly roost in roof spaces and it is therefore vital that contractors are aware of their obligations.

All species of bat and their breeding sites or resting places (roosts) are protected under the Conservation of Habitats and Species Regulations 2017 and the Wildlife and Countryside Act 1981.

It is an offence for anyone intentionally to kill, injure or handle a bat, to possess a bat (whether live or dead), disturb a roosting bat, or sell or offer a bat for sale without a licence. It is also an offence to damage, destroy or obstruct access to any place used by bats for shelter, whether they are present or not.



If bats are discovered during the course of the work, work in that area should be stopped and specialist advice sought. This should include consultation with the client.



For further information refer to Chapter E05 Ecology.

2.10.4 Presence of pigeons

The accumulated droppings of pigeons, if disturbed into airborne dust and then inhaled, can cause severe respiratory problems as well as the possible risk of contracting an acute respiratory disease called Psittacosis. It typically causes flu-like symptoms (fever, headache and muscle aches) that usually appear five to 19 days after exposure, but can lead to severe pneumonia and non-respiratory health problems.

Salmonella may also be present in some bird droppings, and is a bacterial infection that can cause severe diarrhoea, fever and abdominal cramps. It is anticipated that the presence of pigeons will mainly be a problem during refurbishment and repair work, or demolition.

If, during early visits to site, it is evident that pigeons are present, or have been, then measures must be taken to clean up the droppings before work starts, using a safe system of work, and to discourage the return of the birds. Workers with a weakened immune system should not be directly involved in tasks involving bird droppings.

This should be considered as part of the risk assessment process.

CONTENTS

Common access equipment

03

3.1	Introduction	38
3.2	Important points	38
3.3	Ladders and stepladders	38
3.4	Podium steps, mini mobile towers and pop-ups	43
3.5	Lightweight staging and trestle scaffolds	44
3.6	Mobile access towers	44
3.7	Other access equipment	48
3.8	Work at Height Regulations 2005	49
3.9	Supporting information	49

Supporting INFORMATION

GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.



**Scan the QR code to complete a brief survey,
and share your feedback on the course.**

Overview

The purpose of this chapter is to give an understanding of the health and safety requirements when selecting and using common forms of access equipment (such as ladders, podium steps, mobile access towers and hop-ups).

While the chapter provides relevant information, formal and recognised training in the actual access equipment system to be used will be necessary before a person can be regarded as competent.

It is important that anyone reading this chapter has a good understanding of the requirements of the Work at Height Regulations 2005.

3.1 Introduction

There is a huge range of access equipment and solutions available to meet every need of the construction industry.

Access equipment includes items such as hop-ups, stepladders, ladders, podium steps, lightweight stagings and trestle systems, powered access (mobile elevating work platforms) and scaffolding.

These can all provide an easily transported means of access to work at height and they can be used safely in certain conditions. Danger often occurs when access equipment is used for a job for which it is not suitable – this particularly applies to ladders and stepladders.

Access equipment should only be erected, altered, used, dismantled or operated by a competent person, who has received adequate training in systems of work and in any equipment that they are planning to use for access.

In accordance with the Work at Height Regulations, all work at height should be avoided, where possible.

When it cannot be avoided, falls should be prevented by selecting the most appropriate form of access equipment for the particular environment and circumstances.



For information on the working at height hierarchy, refer to Chapter D01 Work at Height Regulations 2005.

3.2 Important points

- When selecting suitable access equipment, the working at height hierarchy should be followed.
- Mobile towers and other forms of access equipment must be subject to a statutory inspection, with a report produced, where appropriate.
- All mobile towers must be properly erected. They must remain stable, be suitable for their purpose and be equipped with toe-boards and guard-rails on all working platforms.
- All personnel involved in the installation and dismantling of mobile towers and access equipment must have received appropriate training for the type of equipment in use.
- Installers and users should be familiar with, and refer to, the manufacturer's instruction manual.



The Health and Safety Executive (HSE) has developed a brief step-by-step guide.

This toolkit has been developed to help workers understand the key issues when working at height and the factors to consider when selecting the most appropriate and safest type of access equipment.

3.3 Ladders and stepladders

Traditionally, ladders and stepladders have been used as the way of getting up to or down from a place of work at height (for example, the means of access to a roof or scaffold).

However, under the Work at Height Regulations, a ladder or stepladder that is being used to carry out an activity (for example, painting a first-floor window) could now be considered as a place of work.

The Work at Height Regulations require that employers give adequate consideration to the safety of the user before selecting a ladder or stepladder as either a means of access to height or as a place of work at height.

In deciding whether or not a ladder or stepladder should be used, the employer must carry out a risk assessment and be able to demonstrate that it is not reasonably practicable to use an alternative, safer means of access and that the risks from using the ladder or stepladder are low.

HSE guidance is that ladders and stepladders should only be used as a place of work when the nature of the work meets the following criteria.

- Is of short duration (up to a maximum of 30 minutes).
- Is of a light (up to 10 kg) nature (requires no heavy lifting, carrying or a destabilising pressure applied by the user or equipment in carrying out the work – minimal manual handling).
- Allows one hand to be available at all times for holding on to the ladder or stepladder.
- Requires nothing to be carried that would cause instability of the ladder, stepladder or user.

Additionally, when stepladders are used as a place of work they should be positioned so that the user faces the work as the stepladder is climbed. A stepladder must not be positioned so that the user is side-on to the work, where the nature of the work would apply a sideways pressure and cause the stepladder to become unstable.

Ladders should:

- be set up at an angle of 75° (1 m out for every 4 m up – a ratio of 1:4)
- be positioned on a firm, level surface
- not be vulnerable to impact by pedestrians or traffic
- be used in accordance with the manufacturer's instructions
- be the right ladder for the job (remember that no single ladder will suit all applications)
- be inspected before use to ensure there are no defects
- ideally, be lashed at or near the upper point of rest by the stiles, not the rungs
- extend at least 1 m above the stepping-off place unless an alternative handhold is provided that enables a safe transfer between the ladder and stepping-off place
- be secured at the bottom or footed* if securing at the top is not possible
- not be rested against fragile or flexible items (such as plastic guttering); a ladder stay or stand-off device must be used, as necessary
- never be painted to an extent that the paint could conceal defects
- be subjected to a schedule of periodic inspections with written inspection reports kept, be individually identifiable and proof of inspection demonstrated
- be visually checked by the user for obvious defects before use and not used if found to be defective. If defective the ladder should be taken out of service and reported.

The working area should also be inspected for the presence of overhead hazards, including power cables.

**HSE research has shown that, to be most effective, the person footing a ladder should stand with both feet on the bottom rung at all times. Even then, footing is not an effective method of stopping a long ladder from slipping sideways. Ideally a ladder will only be footed when it is climbed for the first time for the purpose of tying it off.*



Footing a ladder is the last resort and should be avoided, where reasonably practicable, by the use of other access equipment.



For further information on the performance and effectiveness of ladder stability devices visit the HSE website.

3.3.1 Types of ladder

Standing ladders are a single stage ladder up to 6 m in length.

Pole ladders are a variant of standing ladders, but with the stiles having been made from a long, wooden pole cut centrally to give even strength and flexibility. Lengths can vary up to a maximum of 10 m.

The practice of shortening a pole ladder to fit a particular situation should be discouraged. Care must be taken to ensure that the fabric of the ladder remains stable if a tie wire is removed.

Extension ladders consist of two or three sections coupled together and extended by sliding over or inside each other. Longer multi-stage ladders are extended by means of a rope and pulley. The maximum reach of a single run ladder is 9 m. At this height and above, resting points must be provided.

Stepladders are of various types, have flat rectangular treads and are usually free standing.

Platform stepladders have a built in working platform, and newer models can have wheels for ease of movement, handrails and restraint chains. Due to the provision of a platform, these are usually a better option for work that is regularly carried out at the same height.

Roof ladders are ladders with a ridge hook on the top end for securing over the ridge on the opposite side of a roof (not hung on the actual ridge tiles).

Aluminium and steel ladders are available in various types. Their main advantage is that they are light and weather resistant. They can be prone to slipping at the base if the rubber feet are not properly maintained.

COMMON ACCESS EQUIPMENT

Telescopic ladders are a variation of leaning ladders, but they do not all work the same way. They should always be used, stored and transported with care and kept clean, and it is always important that you read and follow the user instructions provided by the manufacturer.



The Ladder Association, a not-for-profit body dedicated to promoting safe ladder use, has produced a safety guide for users of telescopic ladders - an increasingly popular piece of equipment for working at height - following recent reports of unsafe products in the UK. Visit the Ladder Association website to download a free copy of the guidance.

3.3.2 Classes of ladder (BS EN 131)

Portable ladders, steps, trestles and lightweight staging are covered by BS EN 131.

A **duty rating** indicates their suitability of use.

EN 131 Trade and Industrial. Heavy duty and industrial use (for professional users).

EN 131 Domestic. (For non-professional users.)

In December 2017, the British Standards and regulations for ladders and steps changed to meet criteria set out by the recently-revised European standard for all ladders: BS EN 131 *Trade and Industrial – heavy duty and industrial use* (for professional users) and EN 131 *Domestic* (for non-professional users). The new standards ensure products are built with additional safety standards, such as wider bases and stabilisers on ladders over 3 m in height.

If your existing ladders conform to the previous BS 1129 or BS 2037 standards and are in good condition and inspected regularly then you can continue to use them – there is no need to scrap existing ladders and stepladders that are in good condition. As part of the inspection policy it is advisable to add a note referencing the change so that when new ladders need to be purchased they meet the new BS EN 131 standard.



All ladders should be marked with a unique identification number and the class or duty rating. For more comprehensive guidance refer to LA455 Safe Use of Ladders and Stepladders - a brief guide, the new guidance jointly produced by the Ladder Association and the HSE. Released in July 2021, the guidance replaces the HSE Guidance Document INDG455 of the same name.

3.3.3 Inspection of ladders

Every ladder should be inspected on a regular basis and should carry an identification mark, as detailed above. A written record should be kept of all inspections, defects and repairs.



The frequency of ladder inspections should be determined by considering the use, duration and environment in which they will be used. In addition to pre-use checks, it is common for ladders that are used regularly to be thoroughly inspected every three months.

Ladders should not be used if defective in any way and, if damaged beyond repair, they should be destroyed. Regulation 12 of the Work at Height Regulations requires work equipment (including ladders) to be inspected at regular intervals. It also recommends that all ladders are inspected prior to use. Laddertag systems are available that are in line with the Work at Height Regulations and BS EN 131. The safety tag inserts provide a location to record when the ladder was last checked. Each insert also has a checklist printed on the reverse side.

Some considerations for ladder inspections are shown below.

- There should not be any signs of splitting, cracking, splintering, warping or bruising.
- Metal ladders should not exhibit signs of mechanical damage or chemical attack.
- There must not be any undue wear or movement of the rungs.
- All rungs must be in place.
- Wedges and tie rods must be tight.
- Any metal reinforcements must be positioned correctly.
- Feet must not be split or frayed.
- Plastic inserts must be intact and not be excessively worn.
- There should not be any mud or grease on the rungs. Workers should check that their footwear is clear of mud.
- Wooden ladders should be free from rot.
- Locking mechanisms should be checked. If they are bent or the fixings are worn or damaged the ladder could collapse.
- Any locking bars should be engaged.
- Where repairs are not possible, the report should specify that the ladder should be scrapped.

The condition of any ropes and cords, along with pulleys, hinges and any other fittings, should be checked for fraying to ensure that they are all secure with no sign of damage.

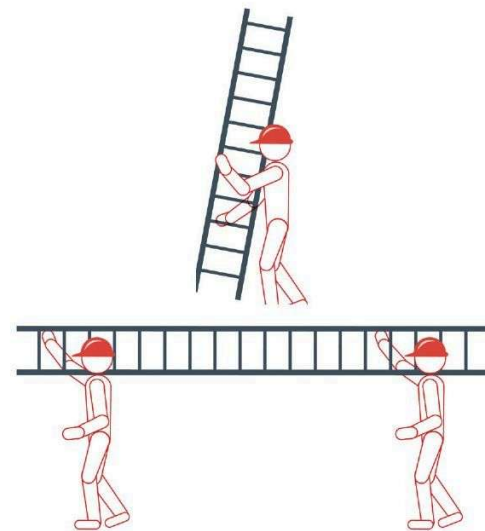
3.3.4 Carrying a ladder

A short ladder may be carried comfortably by having it vertical against the shoulder and holding one of the lower rungs, using the other hand to hold the stile. Longer ladders should be carried horizontally by two people. Care should be taken in negotiating corners and obstacles.

Ladders must not be taken into the vicinity of overhead power lines (OHPL) unless a permit to work has been issued and/or a physical barrier is present, to provide a clearance distance between the OHPL and ladder.

Where any doubt arises the electricity supply company should be contacted for further advice. This may include the installation of temporary, protective overhead line covering (also known as shrouding) or isolation of the OHPL. Even then, extreme care is still advised.

Timber ladders generally do not conduct electricity (unless wet) but aluminium ladders are extremely conductive and are dangerous to use in close proximity to OHPLs.



Carrying a ladder correctly

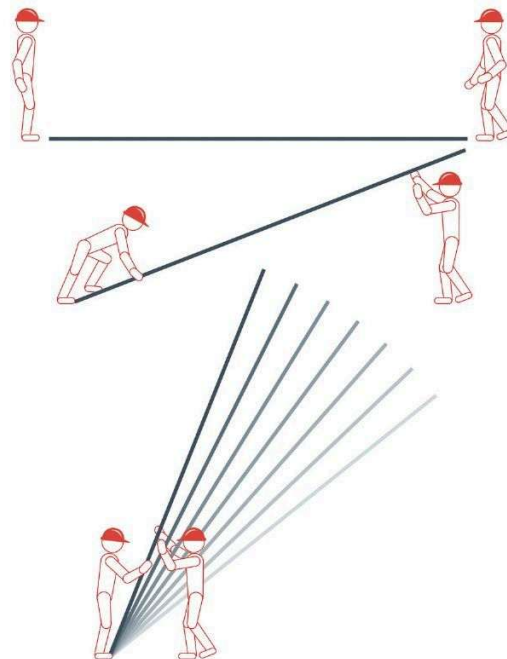
3.3.5 Erecting and lowering ladders

The procedure for erecting a ladder, when the ladder is flat on the ground, is as follows.

- Check there are no overhead obstructions in the vicinity.
- One person stands on the bottom rung while the other takes position at the head of the ladder and takes hold of the top rung, raising the ladder off the ground.
- Then, rung by rung, that person moves towards the foot of the ladder, lifting as they go. The person at the foot grasps the lower rungs as soon as possible and draws the ladder towards them, steadying it at the same time.
- The sequence is reversed when lowering.
- Short ladders may be raised by one person placing the foot of the ladder against a wall or fixture and pushing the ladder upwards, starting at the top, walking, under-running and raising the ladder as they go.

Extension ladders are raised one section at a time and slotted into position. The minimum recommended overlap on extension ladders is shown in the table below. Latching hooks must be properly engaged.

Closed length	Approximate no. of rungs	Overlap of rungs
Under 5 m	Under 18	2
5–6 m	18–23	3
Over 6 m	Over 23	4



How to correctly erect a ladder



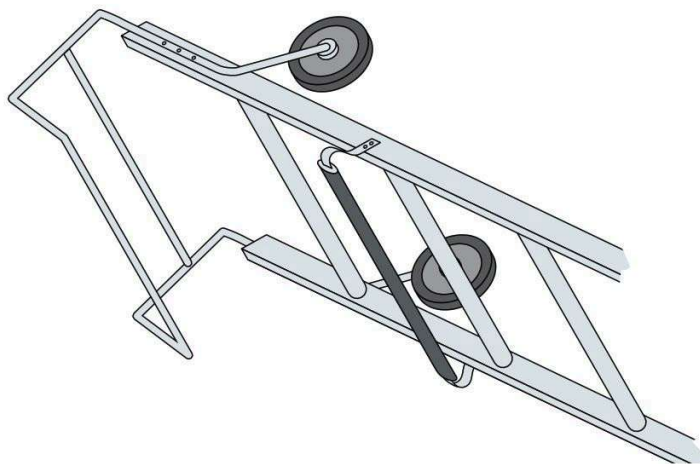
Ladders must never be lashed, tied or spliced together in an attempt to make an extension ladder.

Roof ladders should only be used when it has been established that the use of a safer alternative is not possible. Any work undertaken involving roof ladders must be of short duration and involve only low-risk work. Roof ladders should be erected as follows.

First, the work area must be assessed and, if necessary, an exclusion zone set up with barriers and appropriate safety signs. When all involved are satisfied that the area is segregated and safe, a standing ladder is erected for access to the eaves of the roof.

COMMON ACCESS EQUIPMENT

03



A roof ladder, with ridge/anchor hook (Credit: HSG33: Health and safety in roof work)

It should extend at least 1 m above the eaves of the roof and be properly secured, but not to the guttering, downpipes or any other plastic or fragile material. Proprietary ladder anchors are available and these should be used in preference to other methods.

Adopting good manual handling methods, the roof ladder should then be passed up by an individual at ground level, before being brought up and pushed up the roof on its wheels, with the anchor hook or ridge hook uppermost.

Once over the ridge, the ladder is turned over and the hook engaged. Where possible secure the roof ladder with a rope.

Ideally, where it is necessary to use a roof ladder, access to the lower end of it will be from a working platform at the eaves.

Where access is from a free-standing ladder, the points on ladder safety (refer to 3.3 Ladders and stepladders) must be observed.

Three points of contact should be maintained at all times when climbing the roof ladder.

Roof ladders must be:

- only used by persons who are competent to use them
- positioned to enable easy and safe transfer between:
 - any other ladder used to get to the eaves and the roof ladder
 - the roof ladder and the place of work
- designed for the purpose
- of good construction, strong enough to enable the planned work to be carried out and regularly inspected
- adequately supported to take the user's weight without damaging the roof
- securely fixed to the sloping part of the roof by means of a ridge hook placed over the ridge. Ridge hooks must not bear down on ridge tiles or capping tiles.

3.3.6 Storage of ladders

Ladders, especially if made of wood, deteriorate when exposed to the weather for prolonged periods. Where indoor storage is not available, they should be covered or stored in a protected, well-ventilated position. They should not be exposed to excessive heat, moisture or chemicals.

Ladders should be stored on racks, supported on the stiles only, with sufficient supports to prevent them from sagging. They must not be hung from the rungs or stiles. Aluminium ladders should be kept away from wet lime or cement, as the chemical composition may result in corrosion of metallic components.

Pulleys and hinges should be lubricated and the condition of ropes and cords checked. Any damage or deterioration should be noted and made good before use.

3.3.7 Stepladders

Many of the general rules for the safe use of ladders also apply to stepladders.

In addition, the following points should be considered.

- Stepladders are only to be used based upon a risk assessment confirming their suitability for the task.
- The treads (or steps), hinges, bolts, screws and fixings must all be sound and secure.
- Retaining cords or hinges should be of equal length and in good condition.
- The stepladder must be stable when open and standing on a level base.

- The legs of stepladders should be positioned as far apart as the restraining mechanism allows, with all four legs firmly and squarely on the ground.
- Wherever possible, the stepladder should be positioned so that the person climbing it is facing the work to avoid twisting and unnecessary movement that can create instability.
- Unless the design permits, the knees of the person using the stepladder should be kept below the top step.
- The user should not work from the top three steps of a stepladder (including a step forming the very top of the stepladder) unless there is a suitable handhold.
- Where provided, any gate or guard-rail barrier should be closed when the stepladder is in use.

If it is not practicable to maintain a handhold when a load is being carried, a risk assessment must demonstrate that the use of the stepladder is justified because of the low risk and short duration use of the stepladder.

3.4 Podium steps, mini mobile towers and pop-ups

Podium steps and **mini mobile towers** provide low-level height access, with adjustable height and guard-rails. The use of this type of equipment is preferred to stepladders in most circumstances as it provides a small but stable working platform, complete with guard-rails.

This type of access equipment has the advantage over a stepladder in that it allows the user to work in a safer manner, facing any side of the working platform without it becoming unstable, and it also provides collective means of fall protection.

Podium steps and **mini mobile towers** are lightweight in construction and some types will fold flat for ease of transportation. Specific models are designed to be wheeled through a standard-sized door. Some types of this form of access equipment can be fitted with outriggers to increase stability.

There are many differing types of podium steps and mini mobile towers with different arrangements of use. Some are just wheeled platforms and others have specific installation requirements.



Podium in use, correctly assembled and wheels locked



Example of a push around vertical (PAV)

The instructions for each individual podium or mini mobile tower must be available and understood by the user (for example, the manufacturer's instruction manual) and it is advised (through Regulation 5 of the Work at Height Regulations) that users of equipment of this nature are trained in its use. The Prefabricated Access Suppliers' and Manufacturers' Association (PASMA) low-level access course is one example of suitable training. Users should also ensure equipment is manufactured to BS 8620.

Push around verticals, commonly called **pop-ups**, are an increasingly popular and versatile item of access equipment. These are a category covered by the International Powered Access Federation's (IPAF's) PAL (powered access licence) card. Whilst providing a high degree of safety in most situations, there are a number of specific hazards with the use of this type of equipment and the employer should ensure the following precautions are considered in selecting and using podium steps and mini mobile towers.

- The height of the working platform should be assessed: too low and the user will not reach; too high and the user may be inclined to work from the steps rather than the platform.
- Where adjustable height platforms are in use, ensure that the user is trained in how to adjust the platform and handrails to ensure protection.
- Brakes must be used whilst the platform is in use and users must not be permitted to pull themselves along (surfing) from the top of the podium or mini mobile tower, as this can cause equipment to tip over. Some podiums are fitted with anti-surfing features, intended to ensure users do not try to surf.
- Podium steps and mini mobile towers, like any other access equipment, must have an individual identifying mark and be subject to frequent inspection.
- The manufacturer's instruction leaflet or hire company information leaflet should be available during the inspection to check that the low-level access equipment has been erected in line with these requirements. It is good practice to laminate and attach this information to the item of low-level access equipment being used.

www For further information about low-level access equipment and training visit the PASMA website.

COMMON ACCESS EQUIPMENT

3.5 Lightweight staging and trestle scaffolds

Trestles can be used if the risk assessment shows that the risk of a person falling and injuring themselves is low and the work on top of the trestle is of short duration.

Guard-rails and toe-boards should be installed on all trestle equipment, as per requirements of the Work at Height Regulations.

More modern trestles are similar in many ways to system scaffolding, including the fact that guard-rails and toe-boards are an integral part of the working platform. The spacing of the supports is fixed by the system design, which enables platform boards to fit snugly without overhanging or overlapping.

All lightweight staging should be marked with the maximum permitted distributed loading. This can be done by either specifying the maximum number of persons, allowing for their tools and equipment, or by specifying the maximum safe weight.

The following guidelines should be met.

- Trestles must be set on a firm, level base.
- Only one working platform is installed.
- Guard-rails, barriers and toe-boards are required where a fall would cause a personal injury, as indicated by a risk assessment.
- Trestles should be installed according to manufacturer's recommendations for use. This information should be readily available.
- Scaffold boards used on trestles to form a working platform must be of a consistent length and of equal thickness.
- The trestle assembly must be completely stable when in use.
- A safe means of access to the working platform must be provided (for example, a ladder that is of sufficient length, properly positioned and securely lashed).
- Where locating pins are used, they should be of the correct size and type and not rebar off-cuts or other makeshift items.
- Trestle based working platforms should be subject to a pre-use check. They should also have a formally documented inspection, which can be recorded (for example, on a scaffold tag).



The use of loose scaffold boards, supported on split-head trestles with no means of preventing falls that could cause an injury to the user, is totally unacceptable and should not be considered as an option for working at height.

3.6 Mobile access towers

The use of prefabricated aluminium mobile access towers (mobile towers) on construction sites is a popular alternative to the use of traditional tube and fitting scaffolding.

However, these systems have some limitations and should only be used when they can satisfy both legislative and general site requirements.



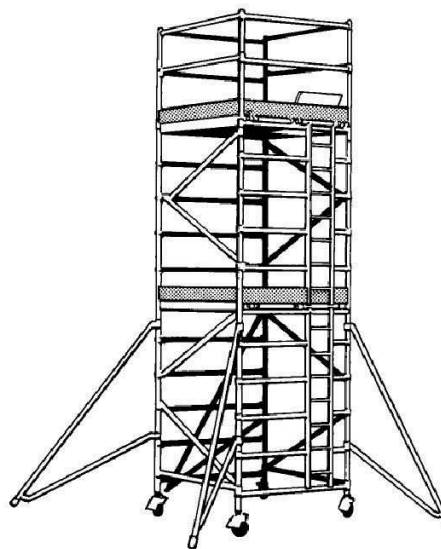
Aluminium and thin-wall steel mobile towers are light, and can easily overturn if used incorrectly.

Five different items make up a basic mobile access tower.

- Frames.
- Braces.
- Platforms.
- Legs.
- Wheels or castors.



For further information on tower scaffolds visit the HSE website.



A prefabricated aluminium mobile access tower

3.6.1 General conditions and provisions

The same points that arise from the Work at Height Regulations in relation to the safe erection and use of system scaffolds apply to mobile access towers but, in addition, the following points should be particularly noted. PASMA has developed two methods of installing guard-rails that do not require the erector to stand on a working platform until the guard-rail frames have been installed.

These are the:

- Advanced guard-rail system.
- Through-the-trap (3T method).

The **advanced guard-rail** system involves the use of specially manufactured, hinged guard-rail frames that can be raised and slotted into position from below.

The progressive erection of guard-rails from a protected area at lower level ensures the operator is never exposed to the risk of falling from an unguarded platform.

The **3T method** involves the erector only passing through the platform trapdoor far enough to be able to locate the guard-rail frames in place.

In many cases, safe erection of the guard-rail sections can be achieved with the erector sitting on the platform with their legs through the trap and their feet supported on the ladder section below.

Both of these methods are accepted by the HSE.



Fall protection equipment must not be attached to aluminium tower scaffolds.

3.6.2 Materials

Mobile access towers may be constructed from steel, but are principally aluminium.

All components must be free from any welding defects, dents, bends or distortion, or any corrosion that could prevent their safe use. Platform boards must be free from holes, cracks, splits or any delamination that would affect their safe use.

3.6.3 Competence

Any person erecting and dismantling a mobile access tower must be competent to do so, having received adequate training (such as that provided by PASMA). Always read and follow the manufacturer's instruction manual and never use the equipment beyond its limitations.

The tower must start off and remain vertical as it is built and used.



For further information about training on the safe use of mobile towers, visit the PASMA website.

3.6.4 Maintenance

Mobile access towers and their individual components are covered by the Provision and Use of Work Equipment Regulations 1998, and must be properly maintained and appropriate to each task.

3.6.5 Preparation and planning

A risk assessment should be carried out to determine whether or not a mobile aluminium tower scaffold is a suitable item of work equipment for the type of work that is to be carried out and the environment in which it is intended to be used.

Factors that should be considered when deciding whether or not it is safe to use a mobile tower are shown below.

3.6.5.1 Ground and surface conditions

Is the surface sufficiently level to use a mobile access tower, if there is no adjustment on the legs for levelling the tower? A mobile tower should not be erected if it cannot be levelled and therefore made stable.

Is the surface strong enough to take the loading? Mobile towers only have a small area of each wheel in contact with the ground and therefore each wheel imposes a high loading at each point of contact.

It may be possible to use sole boards on soft ground to effectively spread the load and allow a mobile tower to be used, providing that there is no chance of the tower sinking, tilting or otherwise becoming unstable.

Are there any features (such as voids, cellars, drain covers, inspection chambers, or signs of tracks or trenches with underground pipes or services) that may not be able to take the direct or indirect loading imposed by one or more wheels?

3.6.5.2 Weather (outdoor use)

Aluminium mobile towers are lightweight structures that can become unstable in moderately high winds.

- Has a check of the weather been made for the period that the mobile tower will be erected? Consider the use of an anemometer (instrument used to measure wind speed).
- Is it possible to tie-in the mobile tower to the structure against which it is to be erected?
- If any doubt exists on the limitations of the mobile tower in high winds, has the manufacturer's or supplier's information been consulted?

COMMON ACCESS EQUIPMENT

3.6.5.3 Overhead power lines

Before starting the job identify any lines that are close enough that make the use of a metal tower unsafe.

Remember, it is not necessary to touch a high-voltage cable to get a shock. Many overhead power lines are not insulated, and the electricity can arc through the air to an adjacent metal object.

Risks from electrocution need to be managed and controlled, especially if a live electrical cable is within a distance of 10 m from the working platform.

If in any doubt, do not erect a tower until such time that the electricity supply company has been contacted and its advice received in relation to clearance distances, shrouding or isolation of the power supply.

3.6.6 Pre-assembly inspection

The competent person who will erect a mobile tower should check that all the components are present and undamaged, are all from the same manufacturer or supplier and are for the same type of tower.

A check should be made that the castors and wheels rotate and swivel freely and that they have a functioning locking device (brake).

3.6.7 Safety during use

3.6.7.1 Stability

Due to their lightweight nature, aluminium mobile towers can have problems with stability – this increases the higher they are built.

Manufacturers provide specific guidance on height-to-base ratios, including circumstances where the installation of outriggers will be necessary. Such guidance should be available to the competent person erecting the tower.

As general guidance, it should be assumed that stabilisers will be required if an aluminium tower is to have a working platform higher than 3 m above ground level.

In addition there are several other factors that can affect stability.

- Ground conditions – bricks or building blocks should never be used to take the weight of any part of the tower.
- Sheeting (outside) – will increase the wind-loading on a mobile tower. In many instances it will not be acceptable to sheet a tower unless it can be tied-in to the structure.
- Overreaching by occupants of the scaffold – if this is so severe that the centre of gravity is moved to a point outside the tower's base area, the tower will overturn. It is far safer to move the tower scaffold.
- Work activity – any work that involves applying a sideways pressure to the adjacent structure (for example, water-jetting at a point that is high on the tower will create an equal and opposite pressure that may overturn the tower. In many cases, tying-in the tower to the adjacent structure may overcome this problem.
- Hoisting materials – if heavy items are hoisted up the outside of the tower, it could become unstable and overturn. Again, tying-in the tower to the adjacent structure may overcome this problem.
- Climbing the tower – access to the working platform should be gained by using the built-in stair or ladder sections. If a vertical ladder is built into an end frame of the tower, the person climbing the ladder must **always** do so on the inside of the tower. Climbing the outside could overturn the tower. Never gain access to the working platform by leaning a ladder against the tower.



Safe working on a mobile tower, with toe-boards and both guard-rails in position and wheels locked



Mobile towers should never be erected to heights above those recommended by the manufacturer.

3.6.7.2 Tying-in a mobile tower

Only special couplers should be used to avoid causing damage to the aluminium tubing. Advice on the horizontal and vertical frequency of ties will be supplied by the manufacturer or supplier or, in the case of substantial or linked towers, the scaffold designer.

3.6.7.3 The working platform

The access hatch to the working platform must be closed as soon as everyone working from the tower is on the working platform.

The working platform must be fully boarded unless a design feature of the tower enables safe access and egress and effective guard-rails and toe-boards to be installed around a partially boarded platform.

The Work at Height Regulations state that a working platform must be of sufficient dimensions for the safe passage of people, plant and materials with due regard to the type of work being carried out. However, BS EN 12811-1 and BE EN 13374 recommend that working platforms on all types of scaffold should not be less than 600 mm wide.

3.6.7.4 Gaining extra height

Placing stepladders and ladders on the working platform of a mobile tower to gain additional height is dangerous and must be prohibited. Adjustable legs are only to be used for levelling, and not to gain additional height. If additional height is needed then a further lift should be added, providing this is within the manufacturer's guidelines relating to height limitations.

In some situations the original equipment might no longer be suitable for the task and an alternative form of equipment will be required.

3.6.7.5 Brakes

The wheel brakes must be locked in the ON position at all times when the tower is not being moved. The lightweight nature of aluminium towers presents the potential for unattended towers to be moved by the wind if the brakes are not applied, particularly where they are used on exposed floor slabs at height.

3.6.7.6 Moving a tower scaffold

The tower must not be moved whilst anyone is on the tower or the working platform. Any item that could fall or cause the tower to be unstable whilst being moved, taking into account the condition of the floor surface, must also be removed.

A tower must only be moved by pushing or pulling at the base. Movement should not take place in windy conditions and surfing is strictly prohibited. Towers in excess of 4 m high should not be moved.

The tower must never be moved by towing it with a vehicle or by a person who is on the platform pulling the tower along using an adjacent structure (surfing).

3.6.7.7 Avoiding collisions

Suitable barriers and warning signs should be erected to prevent people or vehicles from accidentally colliding with a tower, particularly where it is erected in a public place.

The size of safety zone around a tower may vary depending on the nature of the work being carried out above.

3.6.7.8 Loading capacity

The capacity of each platform and tower structure is often shown on labels attached to the frame of the mobile tower, or will be in the manufacturer's assembly guide. Never exceed the recommended loading levels.

3.6.7.9 Incomplete towers

When a tower is in an incomplete or unsafe state, it should not be left unattended without the display of a notice stating 'Scaffold incomplete. Do not use'.

A new inspection and report is not required every time a mobile tower is moved to a new location on the same site. However, if guard-rails or other components have to be removed, to enable the tower to move past an obstruction, then a pre-use check must be undertaken by a trained and competent person to ensure the tower has been reinstated correctly.

3.6.7.10 Protecting the public

Extra caution is required when towers are used in public places.

- Erect suitable barriers and warning signs at ground level to prevent people walking into the tower or the safety zone.
- Remove ladders and prevent access by using physical barriers to prevent unauthorised access if the tower is to remain in position unattended. If this is not practical, the tower should be dismantled.

3.6.7.11 Wind strength

Aluminium structures are vulnerable to the strength of the wind. It is recommended by many manufacturers that if the wind reaches a speed of 17 mph then all work should cease on the tower.

If the wind speed is likely to reach 25 mph, the tower should be tied-in to a rigid structure. If there is a possibility of the wind reaching speeds of 40 mph, the tower should be dismantled. The manufacturer's instructions should always be followed when using any tower in windy conditions.

Operators should be aware of the possibility of sudden high winds or gusty conditions in exposed areas. It must be remembered that wind speeds at high levels are often greater than at ground level. Wind speed can also increase as it funnels and eddies between buildings or other solid structures.



For further information on the Beaufort wind scale refer to Chapter D02 Working at height.

COMMON ACCESS EQUIPMENT

3.6.8 Inspection and reporting

The purpose of an inspection is to identify whether the work equipment is safe to use, and to ensure that any defect or deterioration is detected and repaired. Where the tower constitutes a working platform, it must be inspected by a competent person, as shown below.

- After assembly but before being used for the first time.
- After any substantial addition, dismantling or other alteration and after any event that is likely to have affected its strength or stability.
- Towers with a platform at 2 m or more should be inspected at regular intervals not exceeding seven days since the last inspection.
- Towers with a platform at less than 2 m should be inspected at suitable intervals, depending on the frequency and conditions of use.

For towers with a platform at 2 m or more, a written record must be more than simply a scaffold tag system. A written report must be prepared by the competent person.

A copy of the report should be kept on site, and a further copy be retained, for a period of three months from the completion of the work, by the person on whose behalf the inspection was carried out.

3.6.9 Training

Training for any operation that involves working at height is a requirement of the Work at Height Regulations (Regulation 5 – Competence). Adequate training must be provided for those erecting, dismantling, modifying or inspecting any mobile tower equipment or other equipment used for work at height.



Training courses are available from CITB, as well as from manufacturers and suppliers. Certificates should be provided as proof of training.

3.7 Other access equipment

There is a huge range of access equipment and solutions available to meet every need of the construction industry.

3.7.1 Proprietary access systems

Proprietary access systems are easy to transport between sites, quick to erect and provide good access.

They can be used as an alternative to fixed edge protection, where appropriate for the work being planned.

In some cases proprietary access systems can avoid the need for scaffolding. They are appropriate for short duration, minor work.

3.7.2 Hop-ups

Depending on the results of the risk assessment, hop-ups up to 600 mm high may be suitable for some applications, especially wet trades (such as plastering) that involve constant stepping up and down.

Other solutions with handrails are available, but it is important to balance the risk of falls against the risk of musculoskeletal injuries if other solutions restrict or make movement awkward.



An example of a hop-up



Hop-up type platforms are for low-level access requirements.

3.7.3 Stilts

Fit out and finishing trades (such as dry lining, ceiling fixing and plastering) use professional stilts for low-level access.

These should be designed and manufactured for professional use. However, there is not currently a British or European Standard applicable to these products.

Stilts can be used providing the following requirements are met.

- Following a risk assessment and method statement, and once the system of work has been agreed with the site management team.
- They have been inspected before use and maintained in good order.
- They are used in accordance with the manufacturer's instructions.
- Operatives have received training and can demonstrate that they have the skills and competence to use the stilts safely.
- The floor surface is level and clear from obstructions.

3.8 Work at Height Regulations 2005

These regulations govern all work carried out at height in all industrial sectors and the schedules of the Work at Height Regulations contain a number of important dimensional and prescriptive requirements.

The regulations state that suitable and sufficient measures must be implemented to prevent, so far as is reasonably practicable, any person falling a distance liable to cause personal injury.



For more detailed information refer to Chapter D01 Work at Height Regulations.

3.9 Supporting information

There is a huge range of access equipment available to meet every need of the construction industry, and advice can be sought from a variety of means (such as the Access Industry Forum).

Members of the Access Industry Forum (AIF) are listed below.

- ATLAS – Association of Technical Lightning and Access Specialists.
- BSIF – British Safety Industry Federation.
- EPF – Edge Protection Federation.
- FASET – Fall Arrest Safety Equipment Training.
- IPAF – International Powered Access Federation.
- IRATA – Industrial Rope Access Trade Association.
- Ladder Association.
- NASC – National Access and Scaffolding Confederation.
- PASMA – Prefabricated Access Suppliers' and Manufacturers' Association.
- SA – Scaffolding Association.
- SAEMA – Specialist Access Engineering and Maintenance Association (Suspended access).
- WAHSA – Work at Height Safety Association (Fall protection).

No Falls Foundation



The No Falls Foundation is the first and only UK-based charity devoted exclusively to the work at height sector. With stakeholders including the Ladder Association, the AIF and the HSE, the foundation is dedicated to preventing falls from height and helping people affected by the life-changing consequences of a fall.

For further information on risk assessment, selecting the right equipment and the importance of training, visit the No Falls Foundation website.

